

QMaxent — Manual

Maxent species distribution modeling in QGIS

Byeong-Hyeok Yu

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1. QMaxent Manual

Welcome to the QMaxent user manual. **QMaxent** integrates the elapid Python library into QGIS to deliver the full Maxent species distribution modeling (SDM) workflow — training, spatial cross-validation, jackknife variable importance, projection, and survey planning — through a familiar bilingual interface.

1.1 Where to start

If this is your first visit:

1. Read Introduction for the 2-minute orientation
2. Install the plugin → Installation
3. Set up Python dependencies → Dependencies
4. Download the bundled example data → Example datasets
5. Walk through your first model in 5 minutes → Quick start

If you already use QMaxent and want to dig into a specific feature, the User guide chapters mirror the plugin's five tabs one-for-one. Each can be read on its own.

If you want to see the tool put through its paces on real datasets, the Worked examples cover three case studies of increasing complexity, including a reproduction of the Lee et al. (2025) fairy pitta study published in *Global Ecology and Conservation*.

1.2 Conventions used in this manual

- **Bold** marks UI labels (buttons, tabs, menu items) you click.
- `monospace` marks file paths, code, and parameter values you type.
- Numbered tabs (①, ②, …) refer to the QMaxent Analysis dock tabs.
- Inline citations like *(Phillips et al. 2017)* point to entries in the References chapter.
- Sidebar boxes flag warnings (⚠), tips (💡), and notes worth a second glance — but most of the manual is plain prose; we do not use boxes decoratively.

1.3 Citation

If you use QMaxent in your research, please cite the software using the metadata in the repository's `CITATION.cff`. The same metadata is rendered live on the project home page at osgeokr.github.io/qmaxent and in the References chapter of this manual.

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2. Introduction

QMaxent is a QGIS plugin that brings the full Maxent species distribution modeling (SDM) workflow inside the GIS. From a single docked panel you can choose presence points, register environmental rasters, train a regularised Maxent model, evaluate it with spatial cross-validation, project it across the landscape, and generate field-ready candidate survey sites — without leaving QGIS or writing a line of code.

2.1 What is QMaxent

Under the hood, QMaxent wraps the elapid Python library (Anderson 2023). Elapid implements the maxnet algorithm — a penalised Poisson formulation that is provably equivalent to the original Maxent model on a sufficient sample of background points (Fithian & Hastie 2013) — using the lasso-regularised generalised linear model machinery of glmnet (Friedman, Hastie & Tibshirani 2010). The user-facing model is the same Maxent that the broader SDM literature has used since Phillips, Anderson & Schapire (2006); the difference is the surrounding tooling, not the inferential engine.

2.2 Intended users

QMaxent is for ecologists, conservation planners, and biogeographers who:

- Already use QGIS for spatial analysis and want SDM as an integrated step rather than a separate Java GUI
- Need cross-validated, *spatially honest* performance estimates rather than an over-optimistic in-sample AUC (Roberts et al. 2017)
- Want their results to be **reproducible** — fixed seeds, configuration written to a multi-sheet Excel supplement, model serialised as a portable `.pkl`
- Work bilingually (Korean / English) and need a UI in their language

You do not need to know Python. You do need a working knowledge of vector points and raster layers in QGIS, and enough domain knowledge of your study system to choose appropriate environmental predictors.

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2.3 How it compares to other SDM tools

Tool	Interface	Engine	Status
QMaxent	QGIS plugin	elapid (maxnet, Python)	Active, this manual
Java MaxEnt	Standalone Java GUI	Original Maxent	Stable; broadly used
ENMeval (R)	R package	maxnet / dismo	Strong CV / tuning support (Muscarella et al. 2014)
Wallace	Shiny web app	Multiple (incl. maxnet)	Reproducibility focus (Kass et al. 2018)
dismo (R)	R package	Multiple SDM algorithms	Mature; broad ecosystem

QMaxent occupies the same conceptual niche as the Java MaxEnt GUI for QGIS users, with the added benefit of a modern Python runtime that ships with QGIS, an integrated raster-consistency preflight, and built-in spatial cross-validation. The Pitta nympha worked example reproduces a published Java MaxEnt analysis in QMaxent and discusses where the two pipelines agree or diverge.

2.4 Methodological lineage

QMaxent's defaults follow the recommendations of:

- Phillips, Anderson & Schapire (2006), Phillips & Dudík (2008), and Phillips et al. (2017) for the core Maxent formulation, regularization, and cloglog output transform.
- Elith et al. (2011) and Merow, Smith & Silander (2013) for practitioner guidance on inputs and settings.
- The maxnet R package (Phillips et al. 2017) for the default rule that selects feature classes by sample size. It is a default, not a guarantee of the best model (Merow, Smith & Silander 2013; Radosavljevic & Anderson 2014).
- Roberts et al. (2017) for spatial cross-validation as the default evaluation method.

Every default in the plugin can be overridden in the ② Parameters tab. Full citations are in References.

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3. Installation

QMaxent installs as a regular QGIS plugin. After the plugin itself is in place, a one-time dependency setup downloads the Python libraries Maxent needs into a separate QMaxent folder that does not affect your system Python or QGIS itself. On Linux and macOS this folder is a Python virtual environment; on Windows it is a plain package folder filled by the QGIS Python's own pip (see Dependencies).

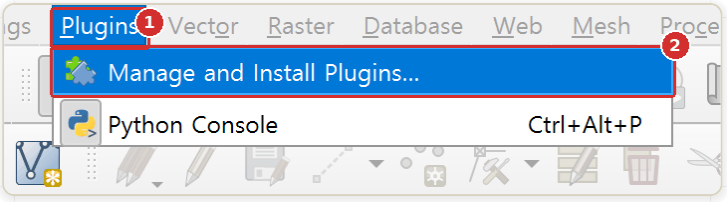
3.1 System requirements

- **QGIS** 3.44 LTR or later
- **Operating system**: Windows 10/11, macOS 12+, or any modern Linux distribution
- **Disk space**: about 600 MB free for the QMaxent package folder
- **Internet access** is required only on first run to download the Python dependencies; routine model fitting and projection run fully offline.

3.2 Install from the QGIS Plugin Repository

QMaxent is published on the official QGIS Plugin Repository.

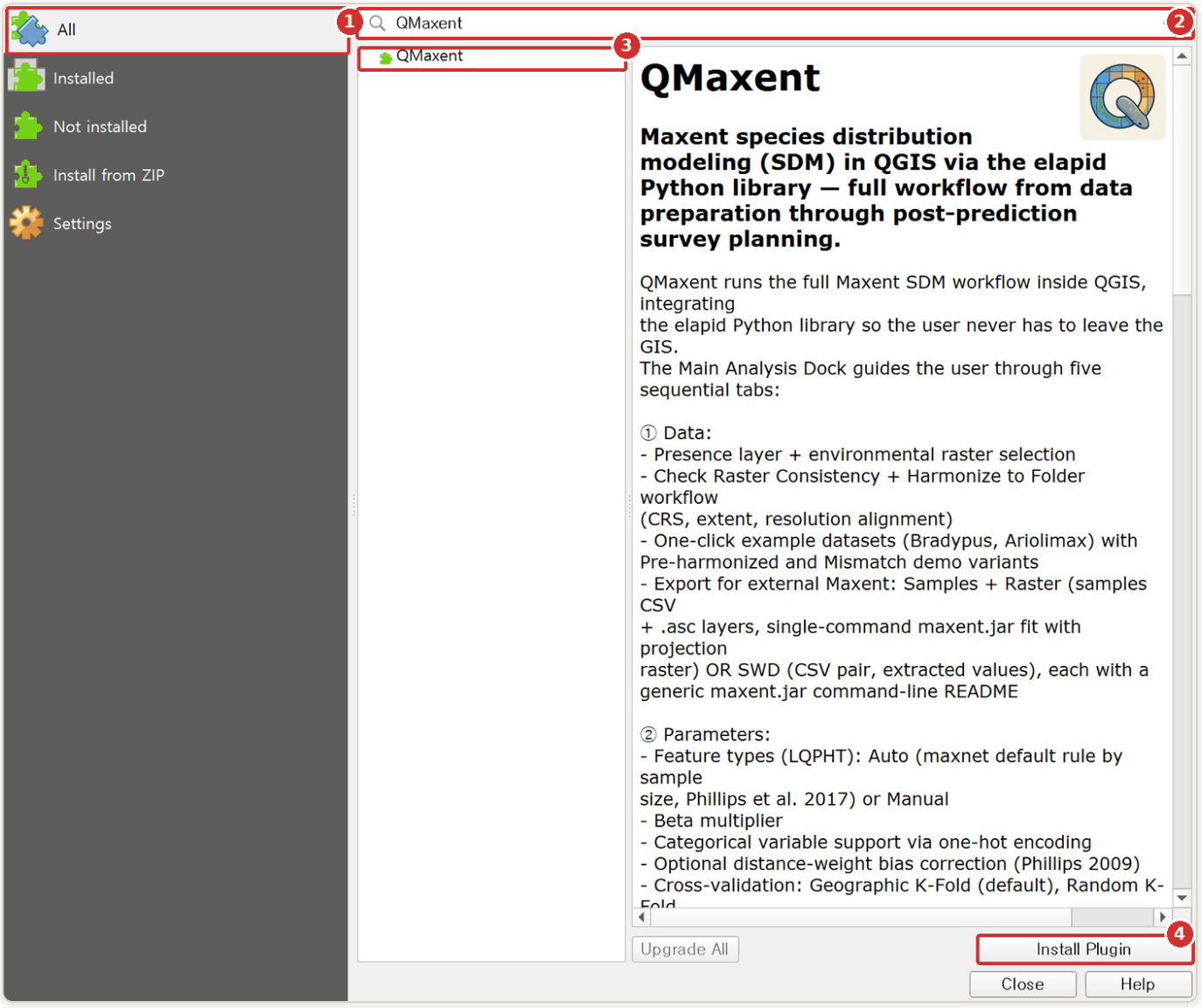
Step 1. Open the Plugin Manager



1. Open the **Plugins** menu.
2. Click **Manage and Install Plugins...**

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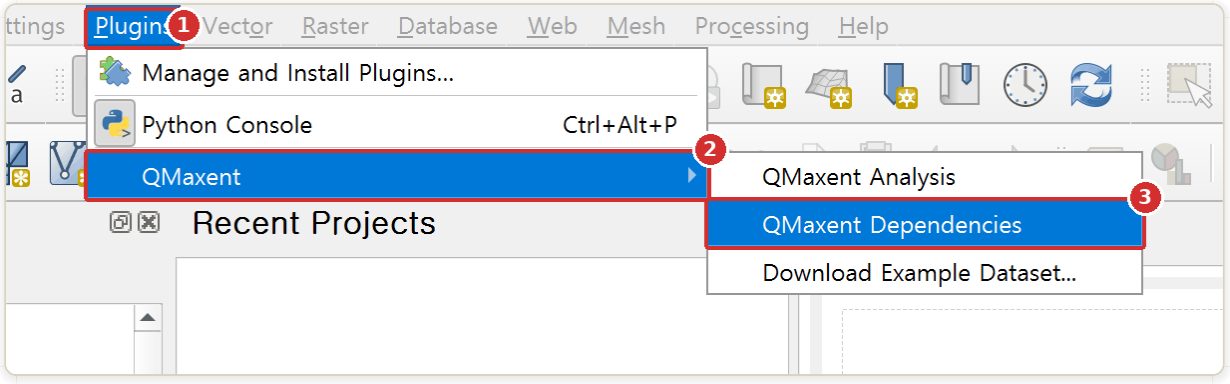
Step 2. Find QMaxent and install it



1. Make sure the **All** page is selected on the left.
2. Type `QMaxent` in the search box.
3. Click **QMaxent** in the result list. Its description appears on the right.
4. Click **Install Plugin**. When the installation ends, click **Close**.

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Step 3. Find the QMaxent menu



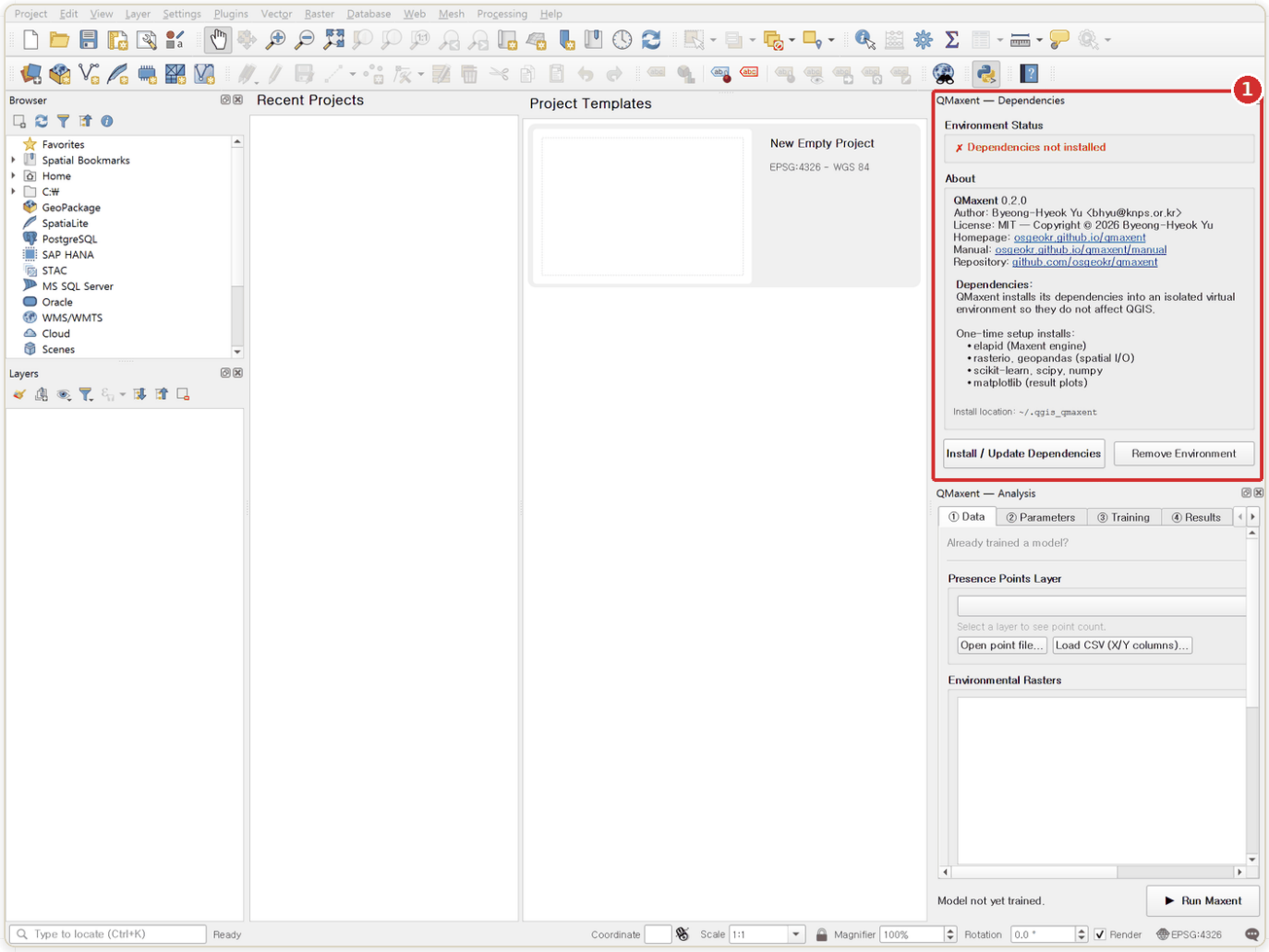
1. Open the **Plugins** menu again.
2. Point to **QMaxent**. Three entries appear.
3. **QMaxent Dependencies** opens the dependency setup described in the next chapter.

Menu item	What it does
QMaxent Analysis	Opens the Analysis dock, where all modelling happens
QMaxent Dependencies	Opens the dependency setup (next chapter)
Download Example Dataset...	Downloads the example data used in this manual (see Example datasets)

3.3 Verifying the installation

Right after the installation, QMaxent opens two panels on the right side of QGIS: **QMaxent Dependencies** at the top and the **QMaxent Analysis** dock below it.

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1. The **QMaxent Dependencies** panel reads *Dependencies not installed* the first time. That is expected: the next chapter installs them.

   If the panels do not appear, choose **Plugins → QMaxent → QMaxent Dependencies**. If that opens, the plugin is installed correctly.

3.4 Updating to a newer version

When a new release ships, the QGIS Plugin Manager will mark QMaxent as upgradable. Click **Upgrade Plugin** in the **Installed** tab. Dependencies generally do not need to be reinstalled across patch releases (0.1.x → 0.1.y). When the package list or the installation method changes, the **Dependencies** dialog reports that the dependencies need updating; click **Install / Update Dependencies** once more. This is the case when you upgrade to 0.2.0 on Windows, because 0.2.0 changed how packages are installed there.

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3.5 Troubleshooting

QMaxent does not appear in the search results

Make sure your QGIS version is 3.44 or later (`Help → About QGIS`). On older versions QMaxent is filtered out as incompatible. Also confirm the **Show experimental plugins** option in **Settings** is set to your preference — QMaxent's stable releases do not require it.

Plugin installs but the menu entry does not appear

Restart QGIS. Some QGIS versions require a restart for top-level menu entries from new plugins to register.

Windows: installation stops with 'ensurepip ... returned non-zero exit status 1'

QMaxent 0.1.x created a Python virtual environment, and this step fails with QGIS 3.44.15 and later on Windows. QMaxent 0.2.0 no longer needs a virtual environment on Windows: it installs the packages with the QGIS Python's own pip into the QMaxent package folder. Upgrade the plugin to 0.2.0 or later and click **Install / Update Dependencies** again. Details are in Dependencies.

Corporate proxy or restricted network

The plugin install itself uses QGIS's network stack and respects QGIS's proxy settings. The dependency installer (next chapter) downloads from PyPI; if your organisation blocks PyPI, ask your network administrator to allow `pypi.org` and `files.pythonhosted.org`.

Once installed, continue to Dependencies.

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4. Dependencies

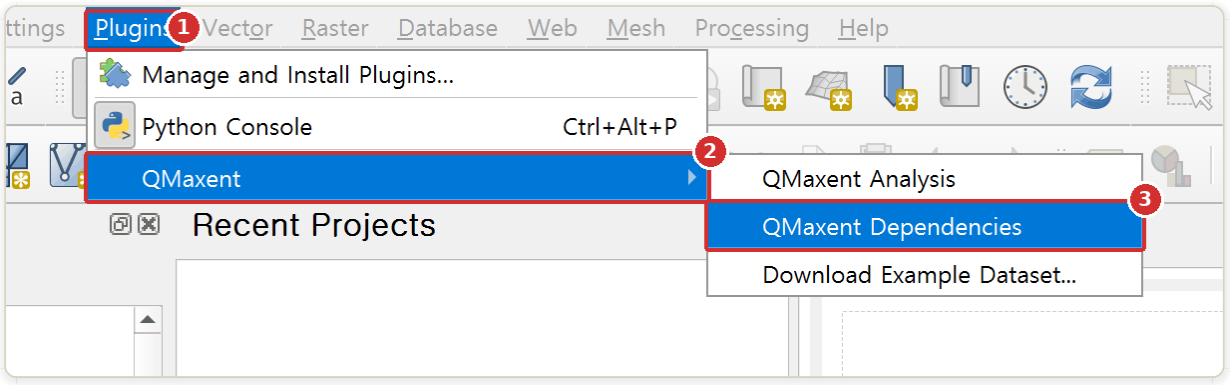
QMaxent's modeling engine — the elapid Python library and its companions (rasterio, geopandas, scikit-learn, matplotlib) — does not ship with QGIS. The first time you use the plugin, you install these dependencies into an **isolated, per-plugin package folder** so they do not affect your system Python or QGIS itself. On Linux and macOS this folder is a Python virtual environment; on Windows it is a plain package folder (see Where the packages are installed).

This is a one-time, ~5-minute setup.

4.1 Step by step

Step 1. Open the QMaxent Dependencies panel

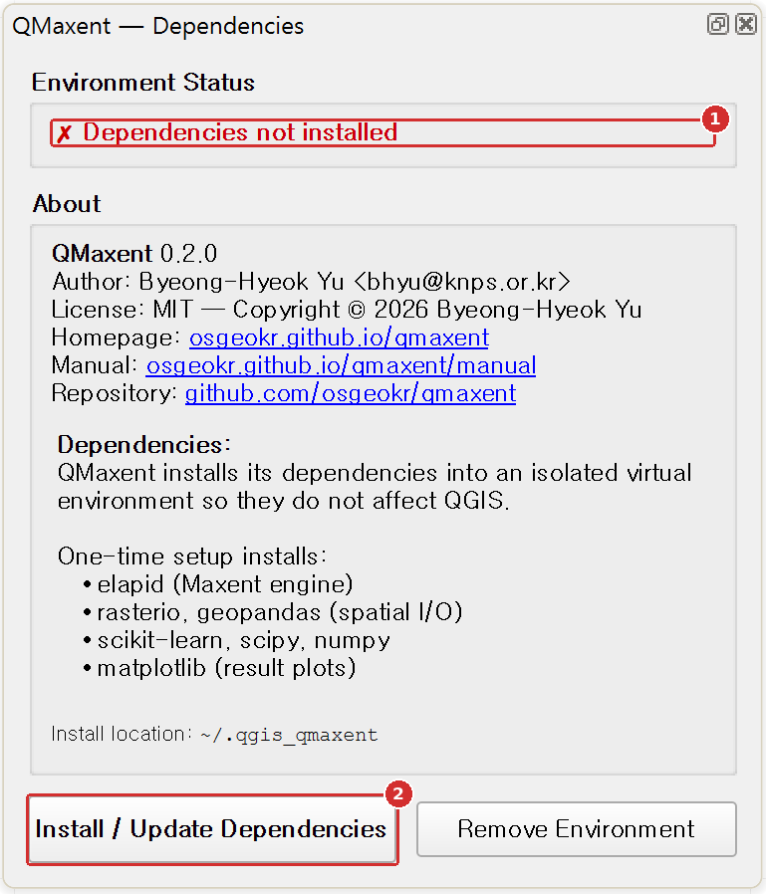
The panel opens by itself right after the plugin is installed. You can also open it at any time from the menu.



1. Open the **Plugins** menu.
2. Point to **QMaxent**.
3. Click **QMaxent Dependencies**.

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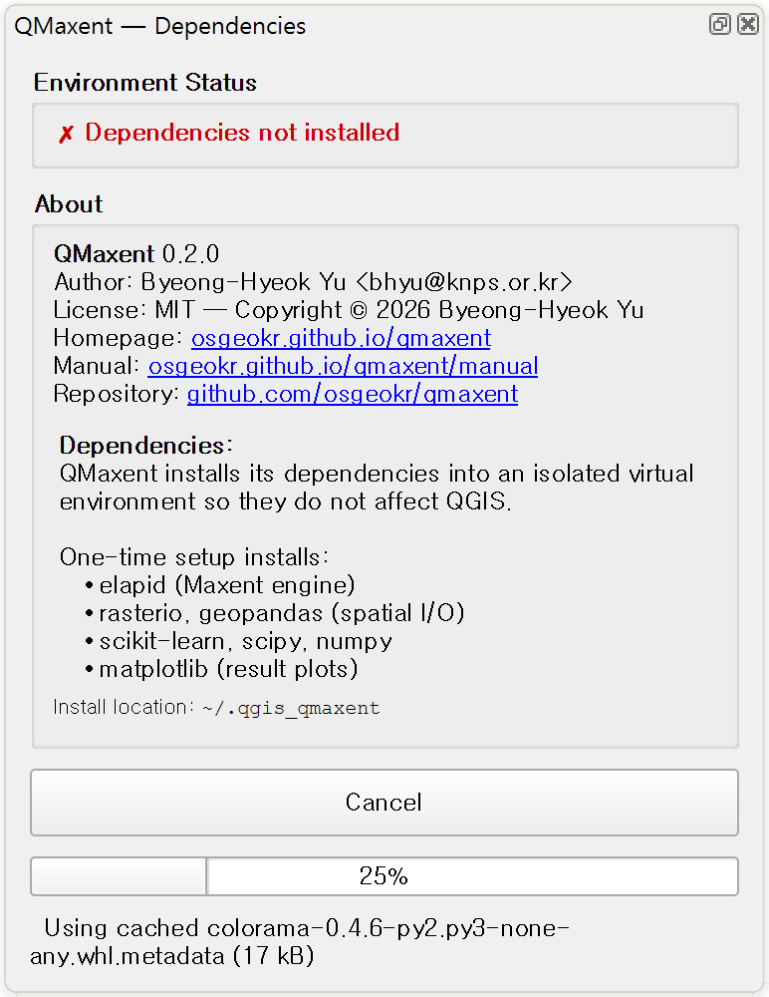
Step 2. Install the dependencies



1. **Environment Status** reads *Dependencies not installed*.
2. Click **Install / Update Dependencies**.

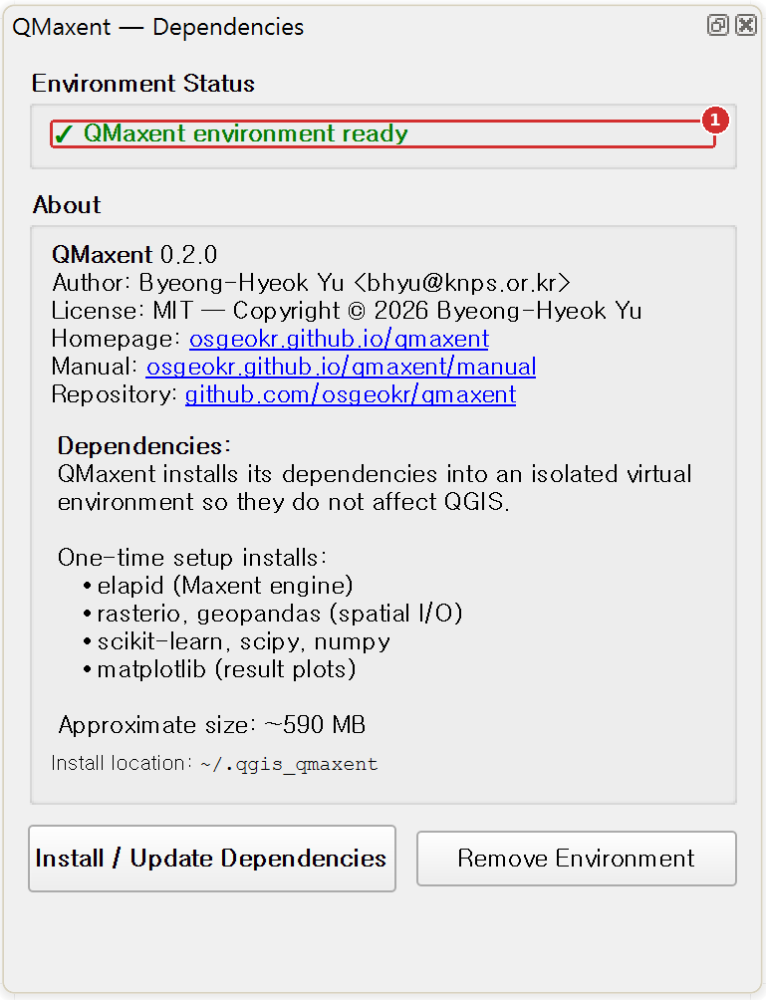
   The packages are downloaded from PyPI into `~/.qgis_qmaxent` (`%USERPROFILE%\.qgis_qmaxent` on Windows), the path shown at the bottom of the panel. The download is about 300 to 500 MB (about 590 MB on disk). Allow 3 to 8 minutes on a typical broadband connection. A progress bar and the package being installed are shown below the button while you wait:

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Step 3. Check that the environment is ready

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1. **Environment Status** turns green and reads ✓ *QMaxent environment ready*. You can close the panel and open **QMaxent Analysis**.

4.2 Environment Status messages

Message	Meaning
✗ **Dependencies not installed**	First-time setup needed
✓ **QMaxent environment ready**	Everything is installed and ready to model
✗ **Dependencies need updating**	The package list or the installation method changed with a plugin upgrade. Click **Install / Update Dependencies** again

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4.3 What is installed

QMaxent installs the following packages and their transitive dependencies:

- **elapid** — the Maxent engine (maxnet algorithm, spatial cross-validation, projection); see Anderson (2023).
- **rasterio** — raster I/O.
- **geopandas** — vector I/O for presence points.
- **scikit-learn** — cross-validation, ROC, AUC; see Pedregosa et al. (2011).
- **scipy** + **numpy** — numerical core; see Harris et al. (2020).
- **matplotlib** — response curves, ROC, jackknife plots.

These are pinned to versions that QMaxent has been tested against; the plugin's release notes record which versions ship with each release.

4.4 Where the packages are installed

All dependencies go into one folder per QGIS Python version, inside `~/.qgis_qmaxent` (`%USERPROFILE%\.qgis_qmaxent` on Windows). The folder name includes the Python version, for example `venv_py3.12` for the Python 3.12 that ships with current QGIS releases.

Platform	How the packages are installed	Package location
Windows	The QGIS Python's own pip installs the packages into the QMaxent package folder (`pip install --target`). No separate Python interpreter is created.	`%USERPROFILE%\.qgis_qmaxent\venv_py3.12\Lib\site-packages`
Linux, macOS	A Python virtual environment (`venv`) is created from the QGIS Python, and its pip installs the packages.	`~/.qgis_qmaxent/venv_py3.12/lib/python3.12/site-packages`

QMaxent only imports these packages inside QGIS, so on Windows it does not need an interpreter of its own. This change was made in 0.2.0: with QGIS 3.44.15 and later on Windows, the virtual-environment approach of 0.1.x stopped with *"ensurepip ... returned non-zero exit status 1"* before any package was downloaded.

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Two environment variables change this behaviour. Set them before you start QGIS:

- `QMAXENT_CACHE_DIR`: use another folder instead of `~/.qgis_qmaxent`, for example when your home directory is not writable or has little free space.
- `QMAXENT_FORCE_VENV=1`: on Windows, go back to the old virtual-environment method of 0.1.x. Use it only for troubleshooting.

4.5 Remove Environment

Use **Remove Environment** to delete the entire QMaxent package folder (the virtual environment on Linux and macOS) from disk. You might do this to free disk space, force a fresh re-install when something is broken, or switch to a different QMaxent version that pins different package versions.

After removal, the status banner returns to *Dependencies not installed* and **Install / Update Dependencies** becomes available again.

4.6 Troubleshooting

Installation fails with a network or SSL error

Most often this is a corporate proxy or firewall blocking PyPI. Ask your administrator to allow `pypi.org` and `files.pythonhosted.org`. The installer respects QGIS's proxy settings, so configuring those in **Settings → Options → Network** usually fixes it.

Windows: 'ensurepip … returned non-zero exit status 1'

You are running QMaxent 0.1.x with QGIS 3.44.15 or later. Upgrade the plugin to 0.2.0 or later, which installs the packages without a virtual environment on Windows, then click **Install / Update Dependencies** again.

Disk space exhausted during install

The installer needs roughly twice the final size as scratch space while extracting wheels. Free at least 1 GB on the drive that holds your home directory before retrying.

"No module named …" after a successful install

Restart QGIS once. Python's import cache occasionally needs a clean process to pick up the new environment.

Plugin starts but Run Maxent button is disabled

Open the **Dependencies** dialog and confirm the status is green. If it is, restart QGIS; if it is not, click **Install / Update Dependencies** again.

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4.7 Security note on the package folder

QMaxent's package folder is populated only from PyPI, by the QGIS Python's pip on Windows and by a standard Python `venv` on Linux and macOS. The plugin does not execute arbitrary code from remote sources at runtime. Saved-model files (`.pkl`) are a separate concern; see Saving and reusing models for the security note on loading pickle files from untrusted sources.

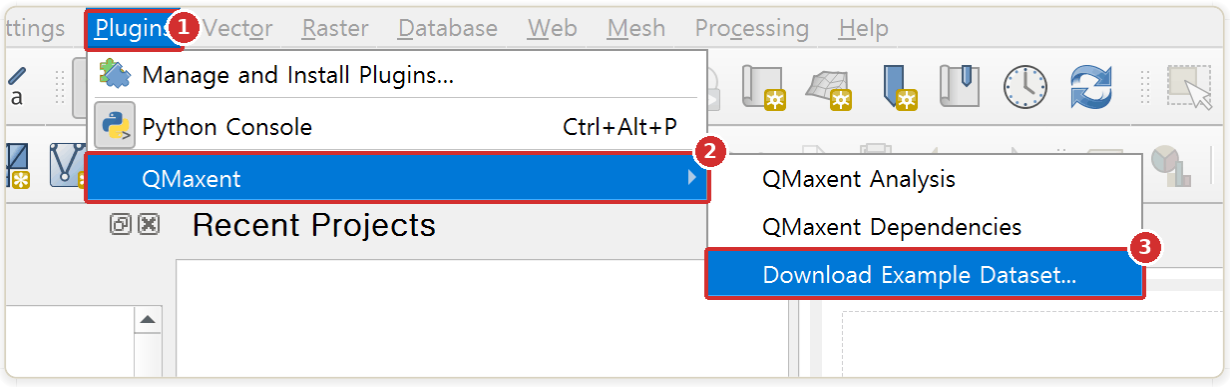
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5. Example datasets

QMaxent ships a one-click downloader for two canonical SDM example datasets, accessed from **Plugins → QMaxent → Download Example Dataset…** They are the same datasets the broader Maxent literature has used for almost two decades, which means you can compare your results directly with published benchmarks.

5.1 Step by step

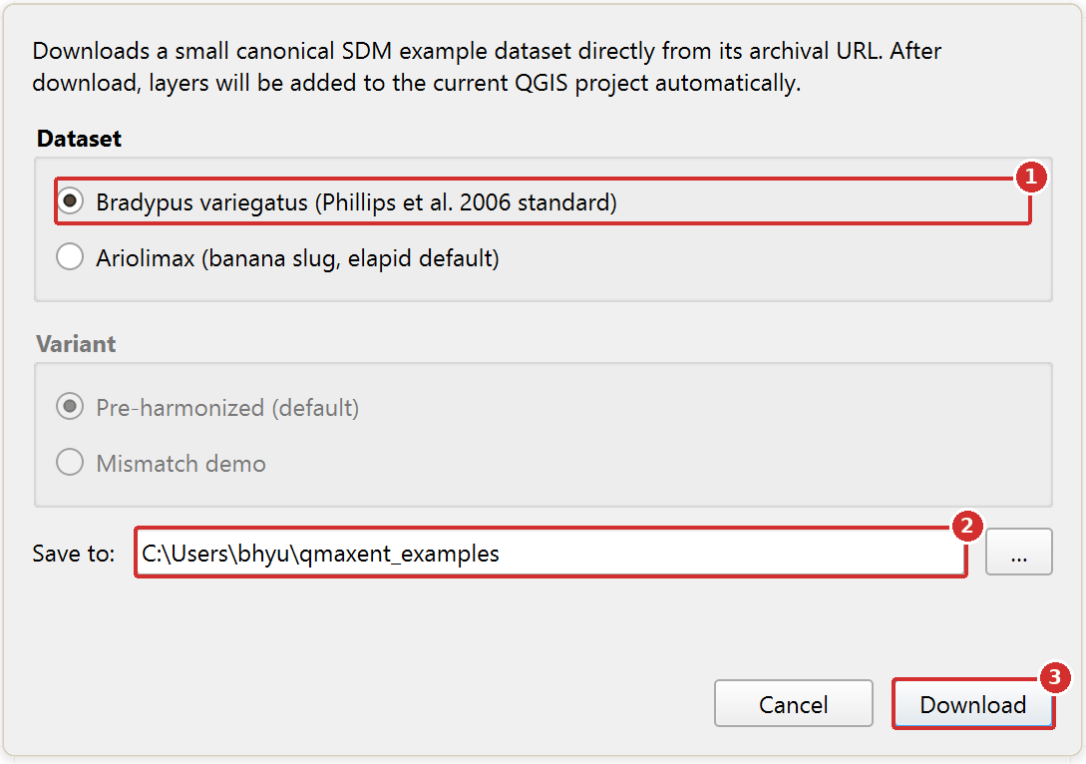
Step 1. Open the downloader



1. Open the **Plugins** menu.
2. Point to **QMaxent**.
3. Click **Download Example Dataset…**

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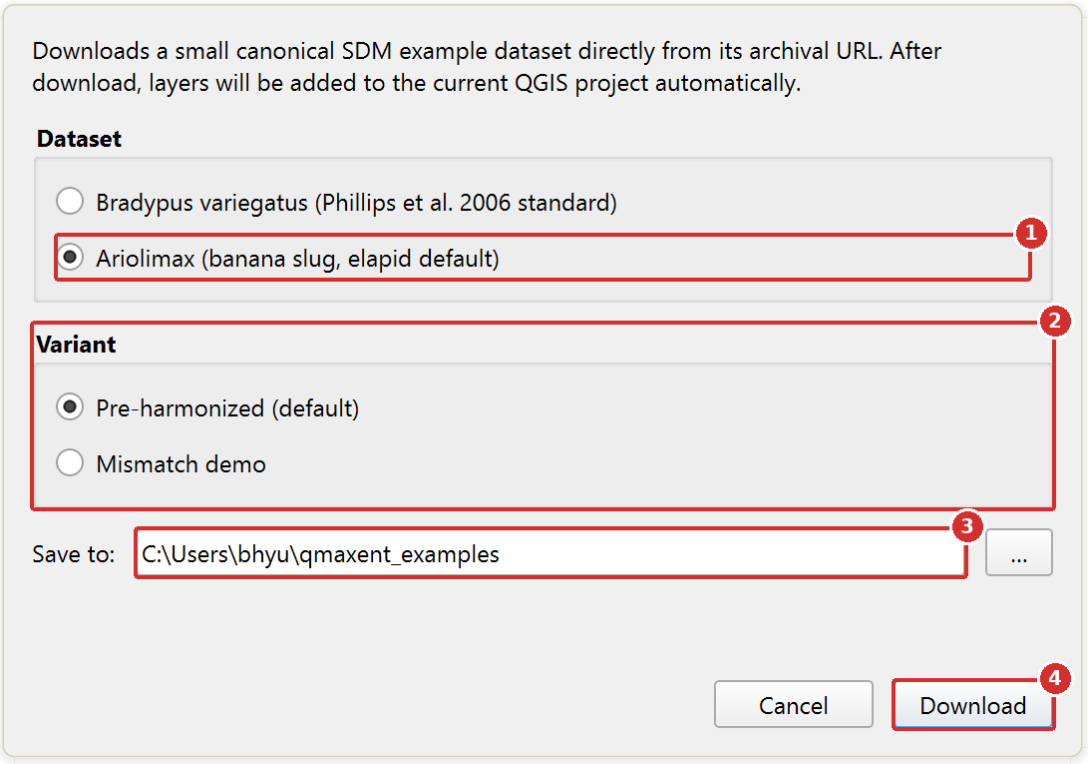
Step 2. Choose a dataset and a folder



1. Choose the dataset. **Bradypus variegatus** is the best one to start with.
2. Check **Save to**. The default is a `qmaxent_examples` folder in your home folder. Click **...** to choose another folder.
3. Click **Download**.

For **Ariolimax**, the dialog also offers a **Variant**:

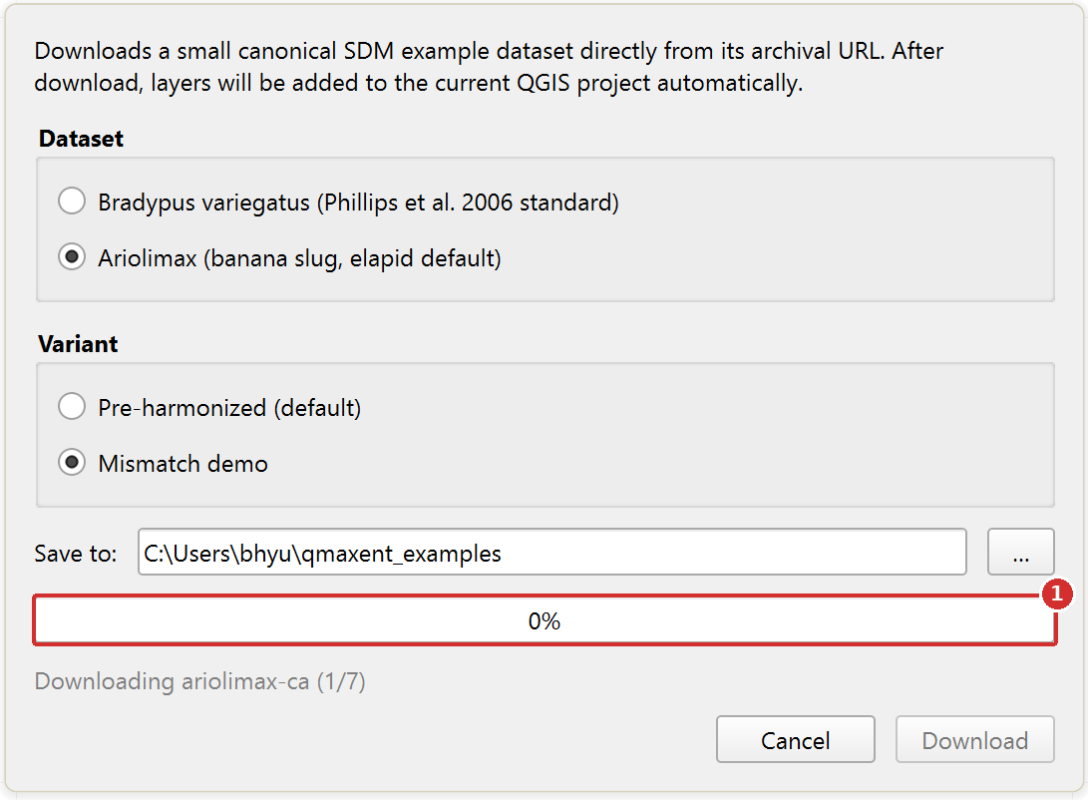
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1. Select **Ariolimax**.
2. **Variant**: **Pre-harmonized (default)** loads the rasters as published, all on one grid. **Mismatch demo** loads copies whose CRS, extent and resolution were changed on purpose, so that you can practise the raster consistency check (see Fixing rasters that do not line up).
3. **Save to** works as for Bradypus.
4. Click **Download**.

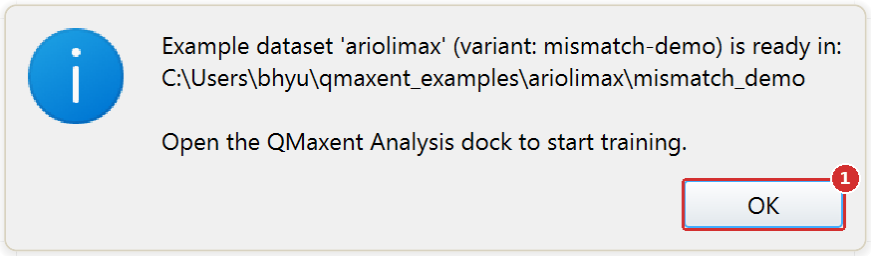
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Step 3. Wait for the download



1. The progress bar and the line below it show which file is being downloaded. Click **Cancel** to stop.

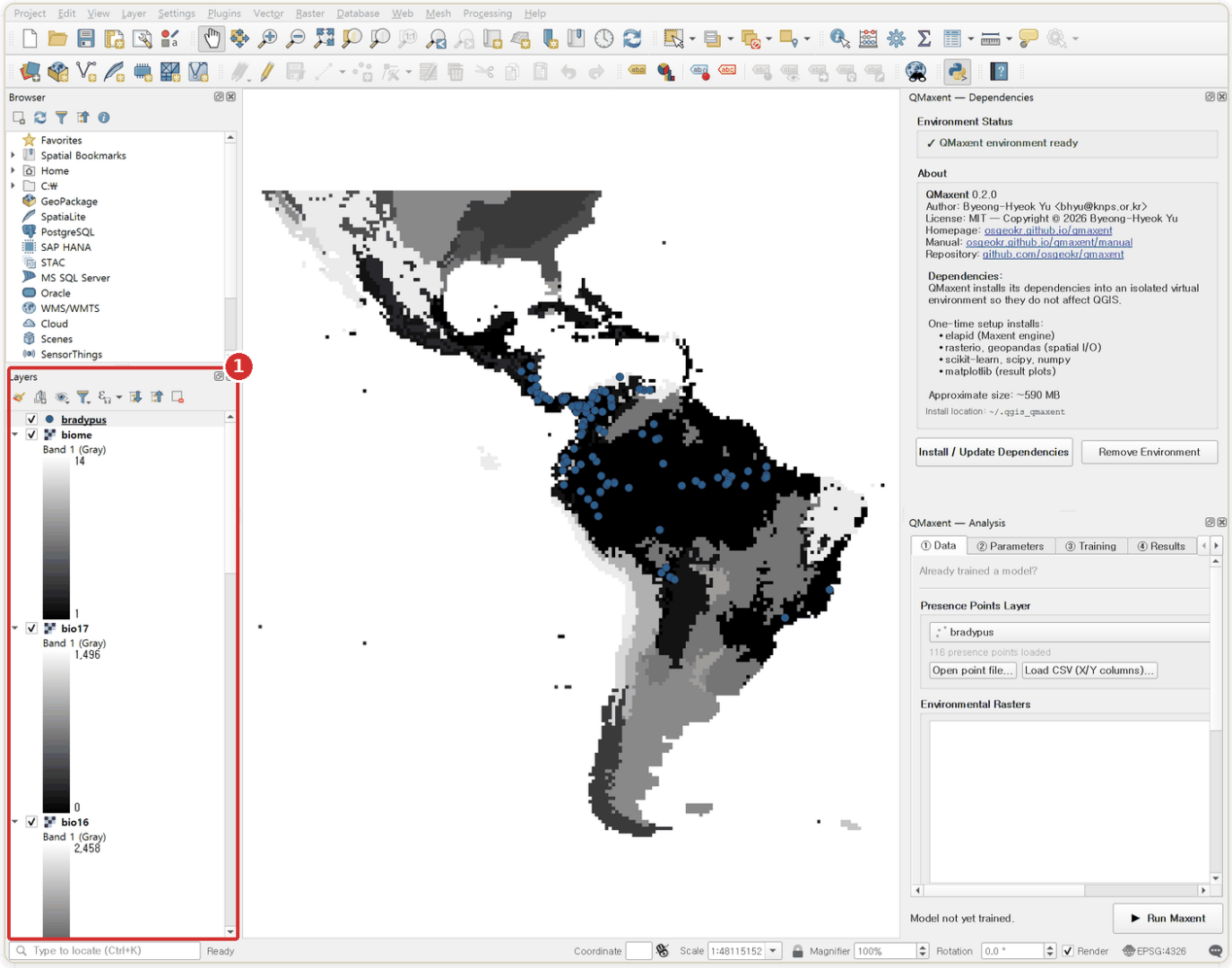
   When the download ends, a message tells you where the files are:



1. Click **OK**.

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Step 4. Check the layers



1. The presence layer and the environmental rasters are added to the **Layers** panel and drawn on the map. You can go straight to the Quick start.

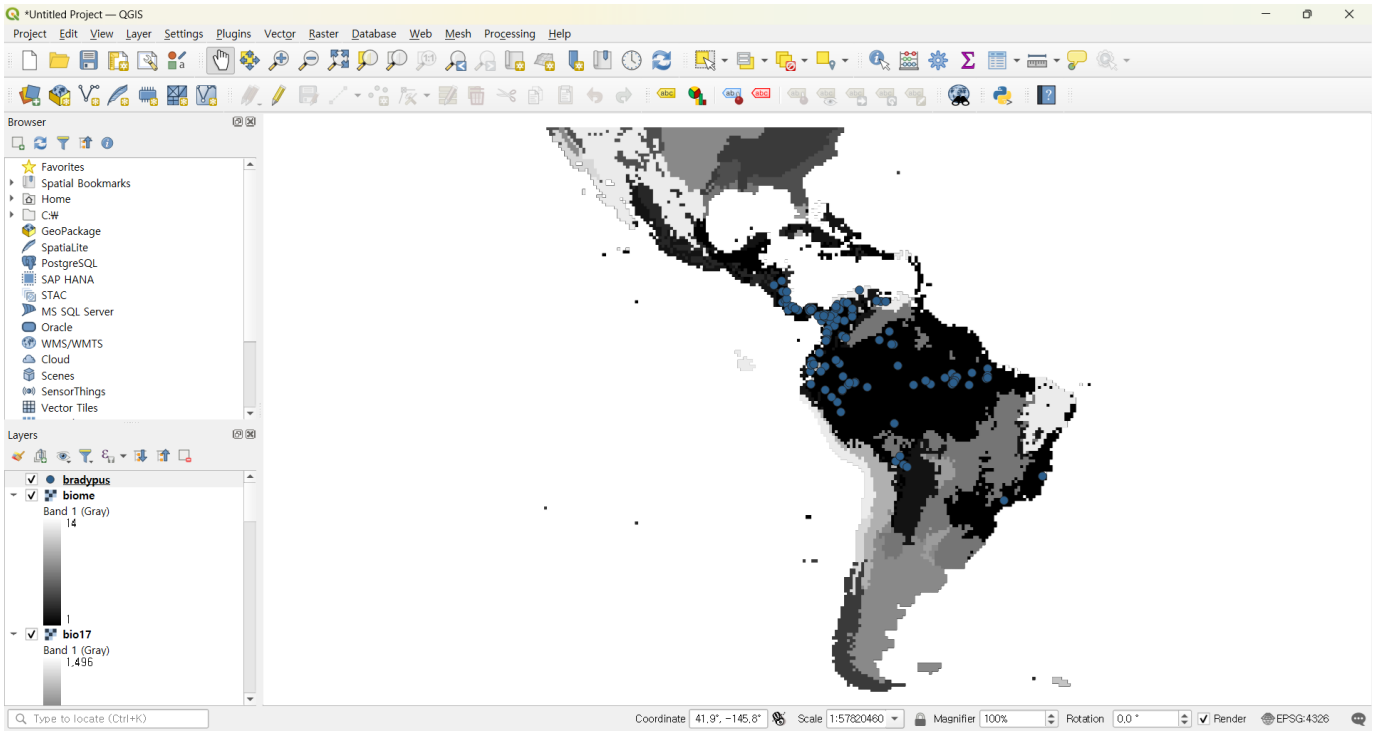
5.2 Bradypus variegatus dataset

The brown-throated three-toed sloth. This is the canonical Maxent test dataset published with Phillips, Anderson & Schapire (2006) and reused in virtually every subsequent Maxent paper.

Layer	Description
`bradypus.shp`	116 presence points across South and Central America
`bio1, bio5, bio6, bio7, bio8, bio12, bio16, bio17`	WorldClim bioclimatic variables (continuous)
`biome`	Biome classification (categorical)

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After download the layers appear immediately in the QGIS canvas:



The dataset is small (well under 100 MB) and the resulting Maxent runs typically finish in under 30 seconds, so it is ideal for first runs and for working through this manual.

5.3 Ariolimax dataset

The Pacific banana slug. This is the default test dataset that ships with elapid.

> **Pre-harmonized or Mismatch demo?**
>
> The published Ariolimax rasters already share one CRS, extent and resolution, so **Check Raster Consistency** passes at once. Choose **Mismatch demo** in the **Variant** box when you want to see and fix a grid mismatch, as in the Ariolimax worked example. The demo copies are written to a `mismatch_demo` subfolder; the original files stay unchanged.

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5.4 Save location and project structure

Whichever dataset you choose, the downloader creates a sub-folder named after the species:

```
qmaxent_examples/
├── bradypus/
│   ├── bradypus.shp     (and .dbf, .shx, .prj, .cpg)
│   ├── bio1.tif         …
│   └── biome.tif
└── ariolimax/
    ├── ariolimax.shp
    └── …
```

Re-running the dialog with the same Save-to path overwrites previously downloaded files, so it is safe to use as a "reset to defaults" button if you want to start a tutorial cleanly.

5.5 Loading the layers manually

The plugin auto-adds layers, but if you ever need to reload them by hand, just drag the `.shp` and `.tif` files from QGIS's **Browser** panel into the **Layers** panel — exactly as you would for any other dataset.

5.6 Where the data comes from

Both datasets are pulled from their canonical archival URLs at download time; QMaxent does not ship them inside the plugin itself. This keeps the plugin package small and ensures you always get the upstream-canonical version. If you need offline access, run the downloader once on a connected machine and copy the `qmaxent_examples` folder onto offline workstations.

Copies of both datasets are also kept in the `example_data` folder of the QMaxent repository (about 15 MB). The files are unchanged copies of the originals, and `README.txt` lists their sources and citations. Download the files and add the layers to a QGIS project.

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6. Quick start

This walkthrough takes you from a freshly installed plugin to a finished habitat-suitability map and a list of field survey sites for *Bradypus variegatus*, the brown-throated three-toed sloth. It is the same species the original Maxent paper used (Phillips, Anderson & Schapire 2006).

Every step shows the real QGIS screen. The red boxes and numbered circles on each picture match the numbered instructions below it, so you can follow along click by click.

Before you start

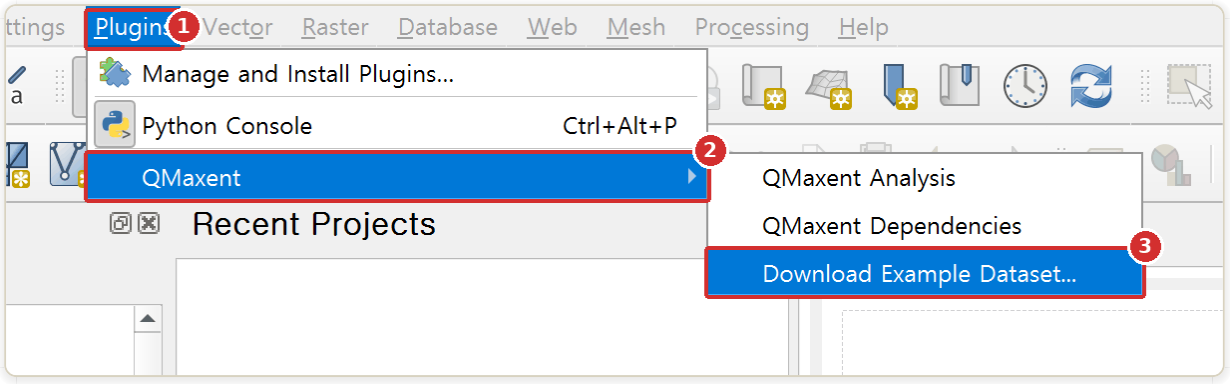
Complete Installation and Dependencies first. The **QMaxent environment ready** status must be shown in green before you continue.

How long does it take?

About 10 minutes in total. Most of that is the model fit in Step 9, which runs in the background while QGIS stays usable.

6.1 Part 1. Get the example data

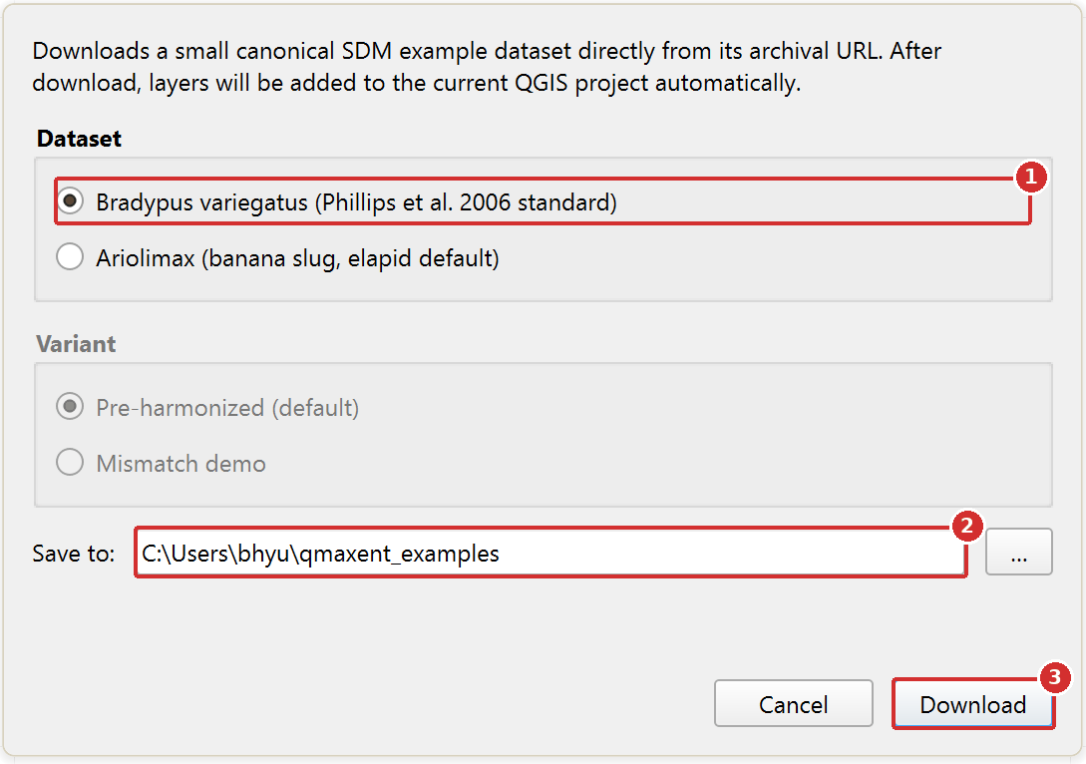
Step 1. Open the example dataset downloader



1. Open the **Plugins** menu in the QGIS menu bar.
2. Point to **QMaxent**.
3. Click **Download Example Dataset...**

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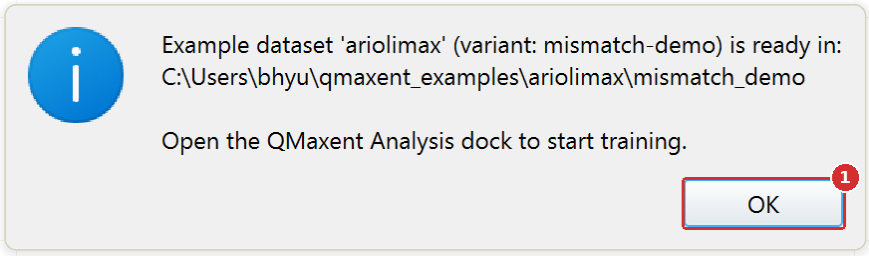
Step 2. Choose Bradypus and download



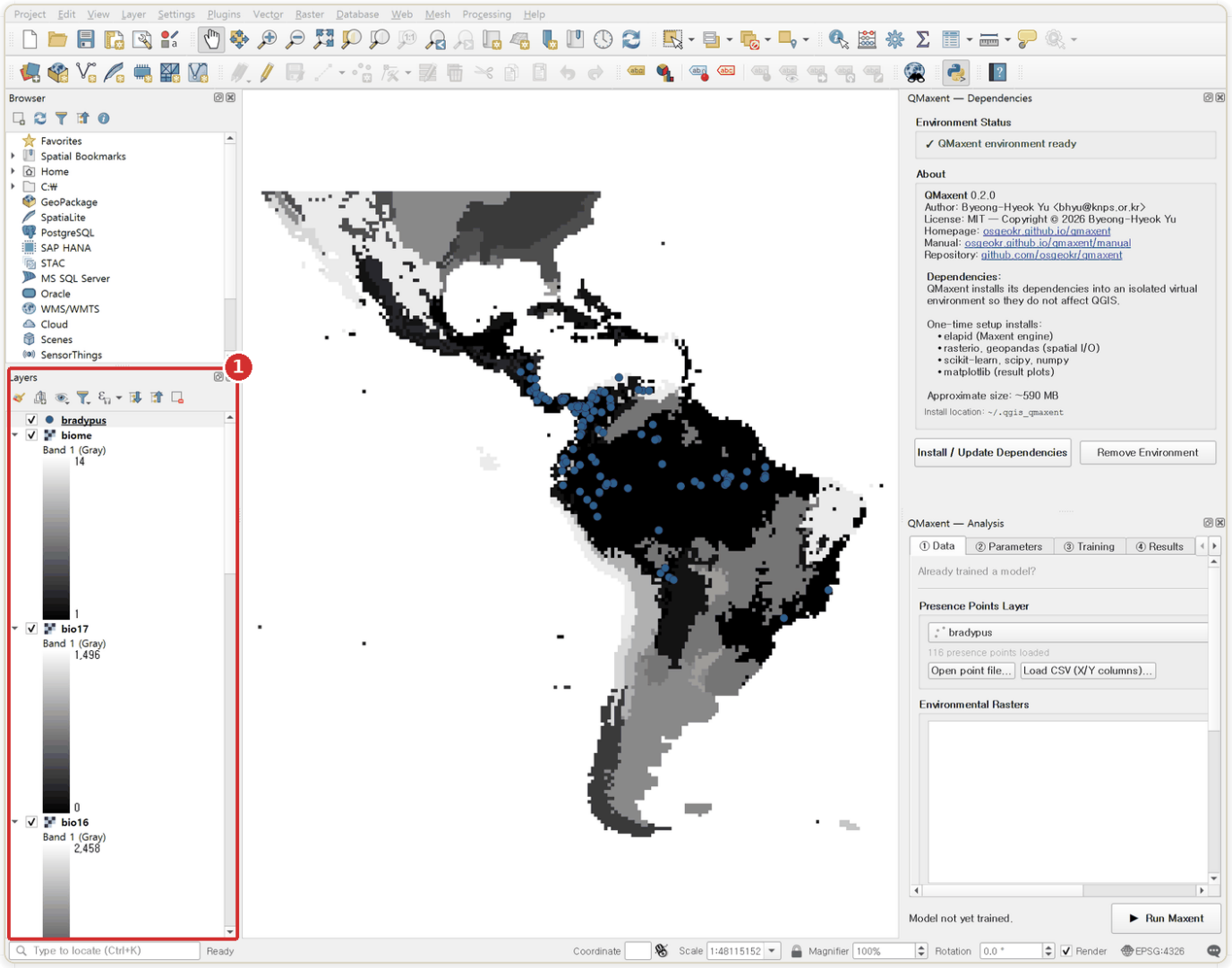
1. Select **Bradypus variegatus**.
2. Check the folder in **Save to**. The default is a `qmaxent_examples` folder in your home folder. Click **...** to choose another place.
3. Click **Download**.

The download takes a few seconds. When the message below appears, click **OK**.

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Step 3. Check the layers on the map



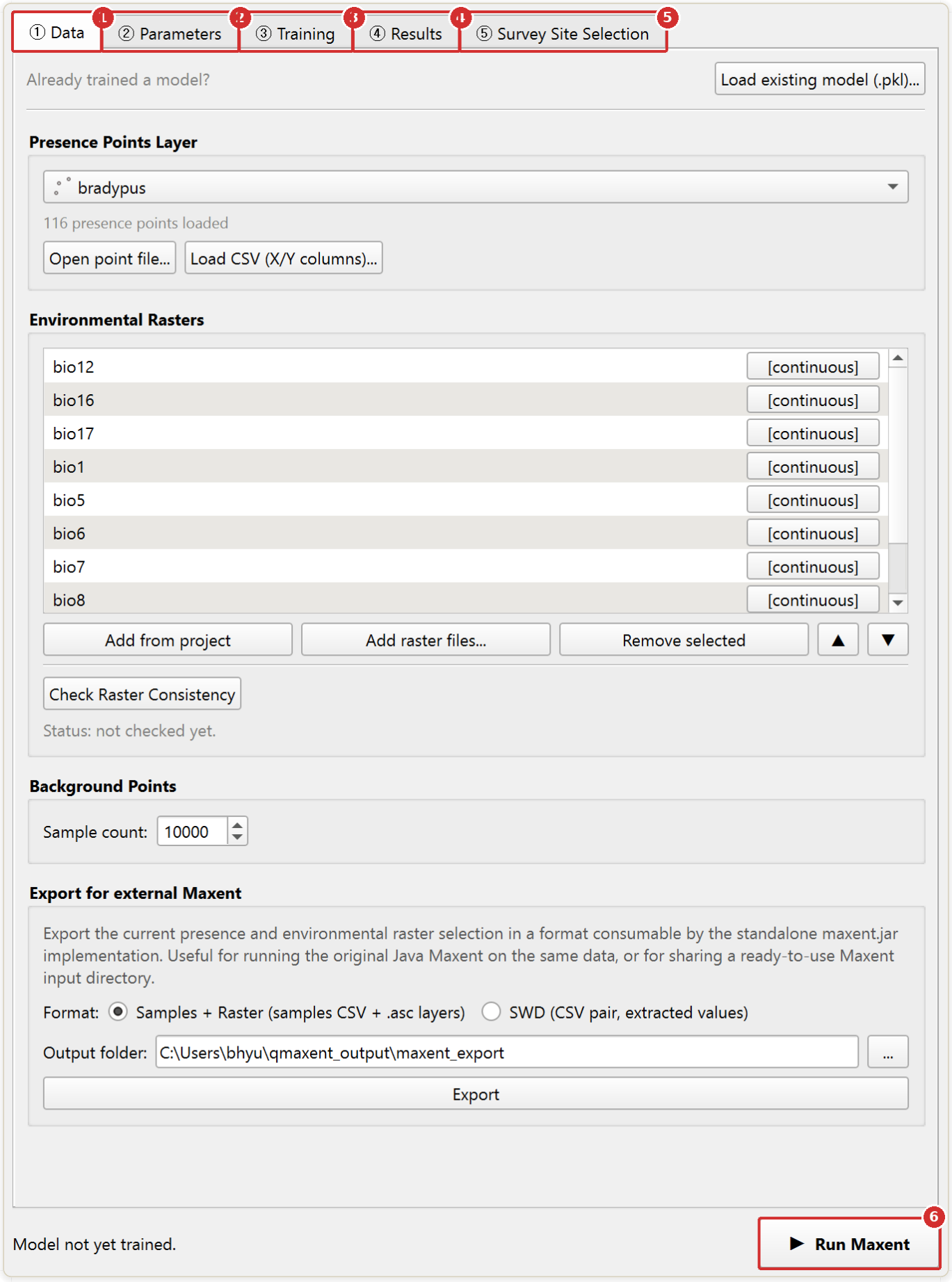
1. The **Layers** panel now lists the `bradypus` presence layer and the nine environmental rasters (`bio1` to `bio17` and `biome`). The blue points on the map are the 116 places where the sloth was recorded.

6.2 Part 2. Prepare the data (① Data tab)

Step 4. Open the Analysis dock

Choose **Plugins → QMaxent → QMaxent Analysis**. The Analysis dock opens on the right side of QGIS.

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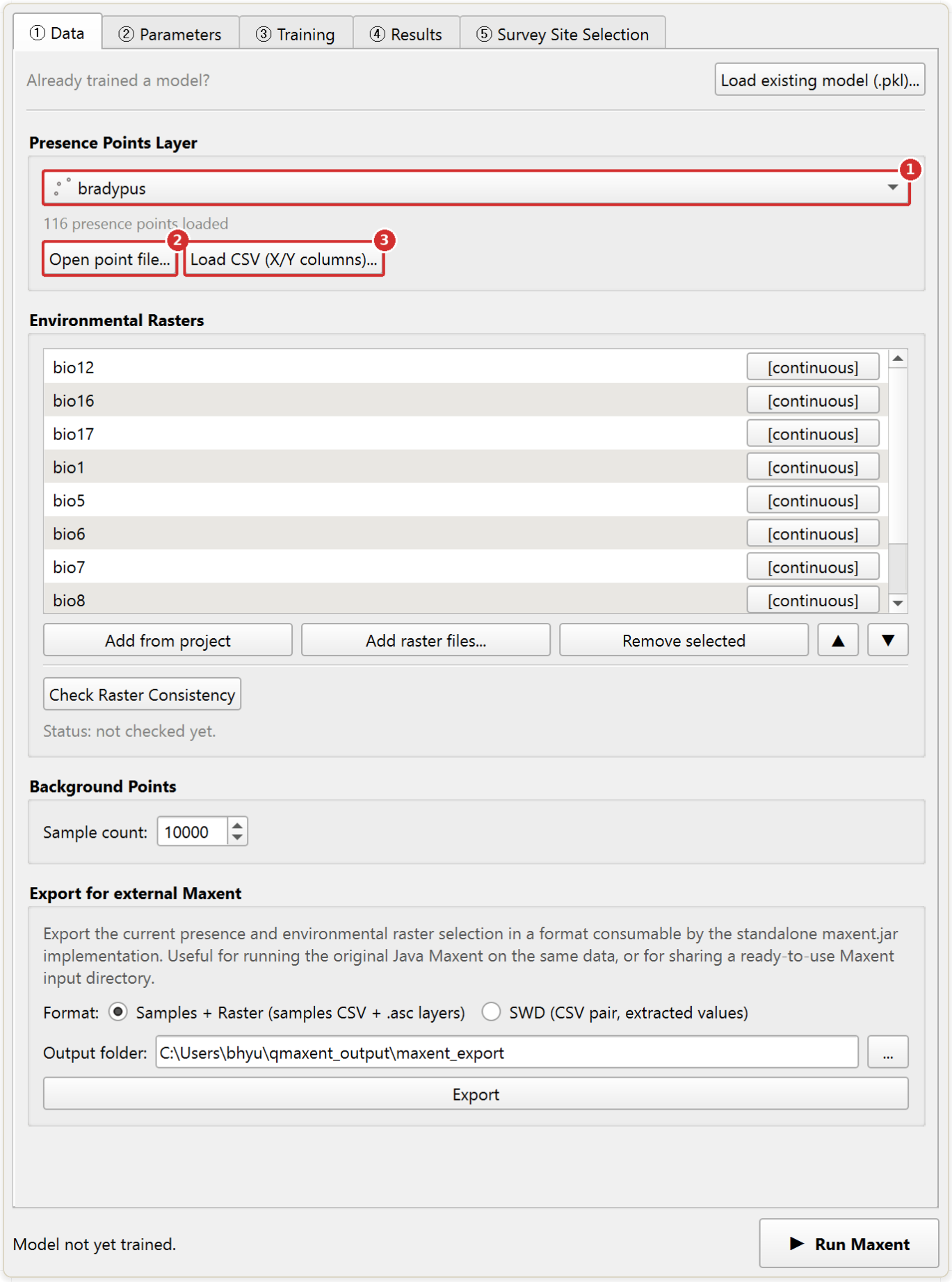
1. ① **Data**: choose the presence points and the environmental rasters.
2. ② **Parameters**: set how the model is fitted and evaluated.
3. ③ **Training**: follow the model fit and read the log.
4. ④ **Results**: look at response curves and variable importance, and make the suitability map.
5. ⑤ **Survey Site Selection**: pick field survey sites from the map.
6. ▶ **Run Maxent** starts the model fit. You use it once the first two tabs are ready.

Work through the tabs from left to right.

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Step 5. Choose the presence points

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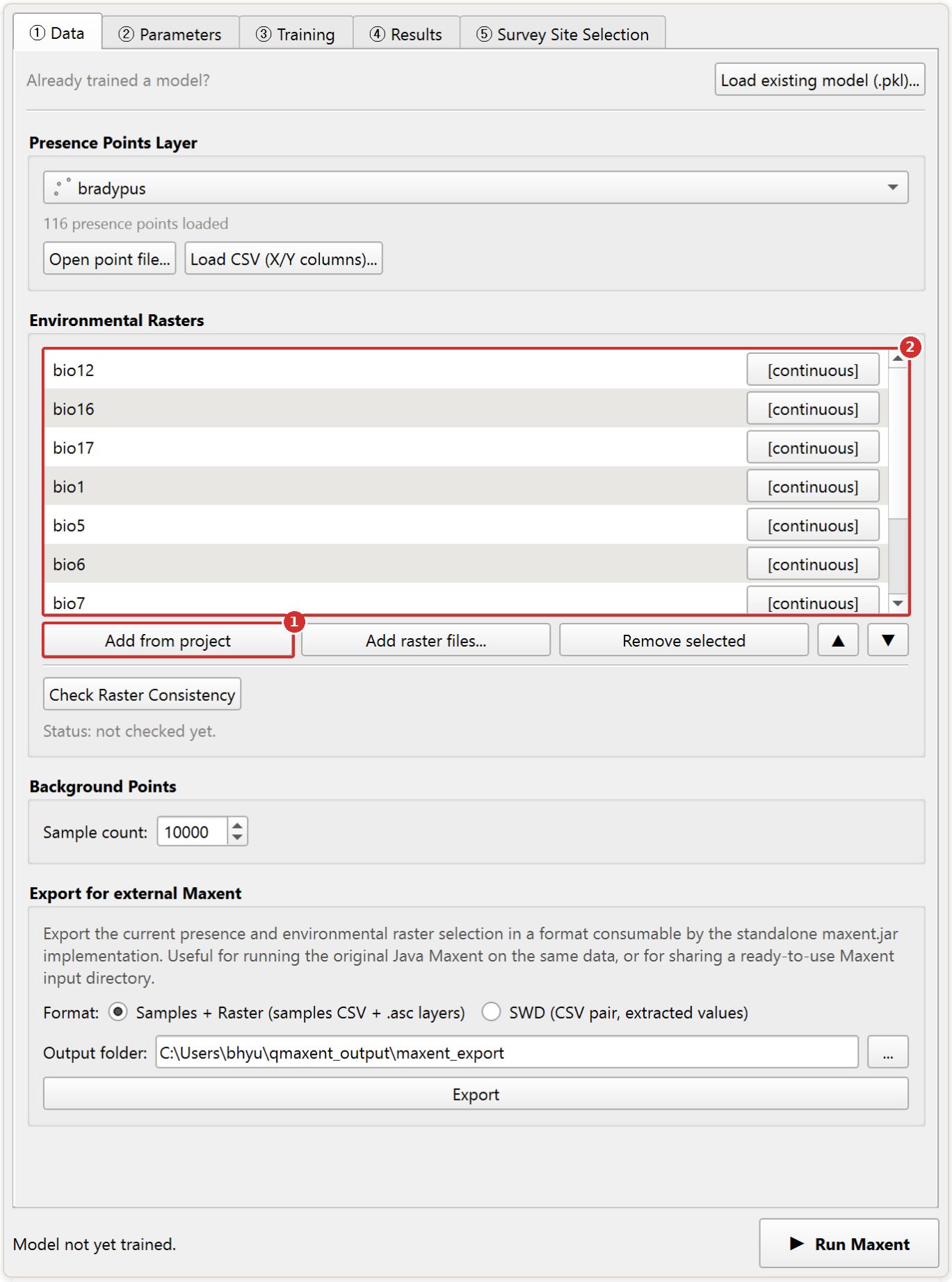
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1. In **Presence Points Layer**, choose `bradypus`. The line below it reads `116 presence points loaded`.
2. **Open point file…** loads a point file (GeoPackage, shapefile and others) that is not yet in your project.
3. **Load CSV (X/Y columns)…** loads a table with coordinate columns. You do not need either button for this example.

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Step 6. Add the environmental rasters

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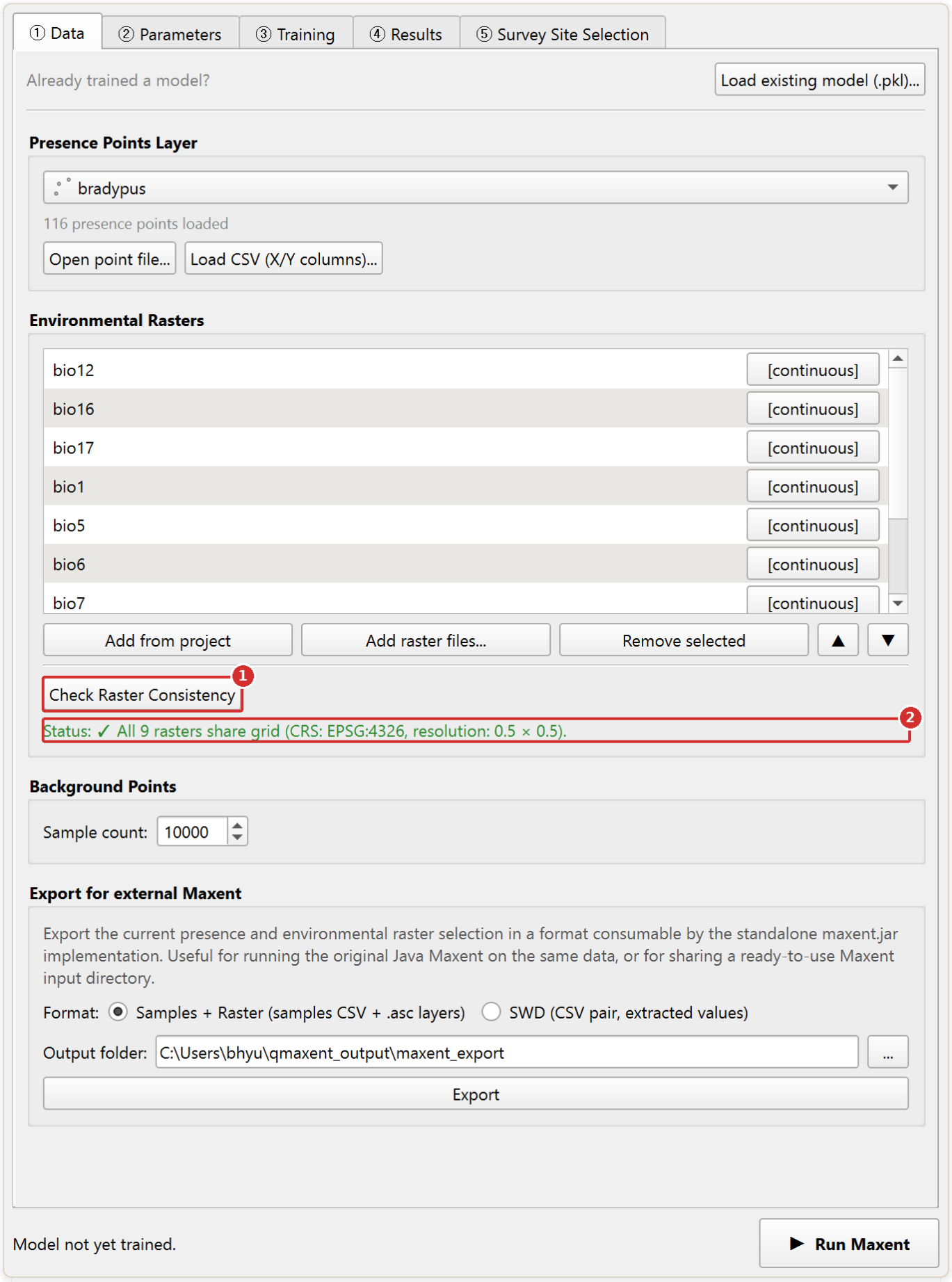
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1. Click **Add from project**.
2. Every raster in the project appears in the list. The button on the right of each row shows whether QMaxent treats the raster as `[continuous]` or `[categorical]`. Click it to change the type. Scroll down the list and check that `biome` is marked `[categorical]`, because its values are class codes, not amounts.

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Step 7. Check that the rasters line up

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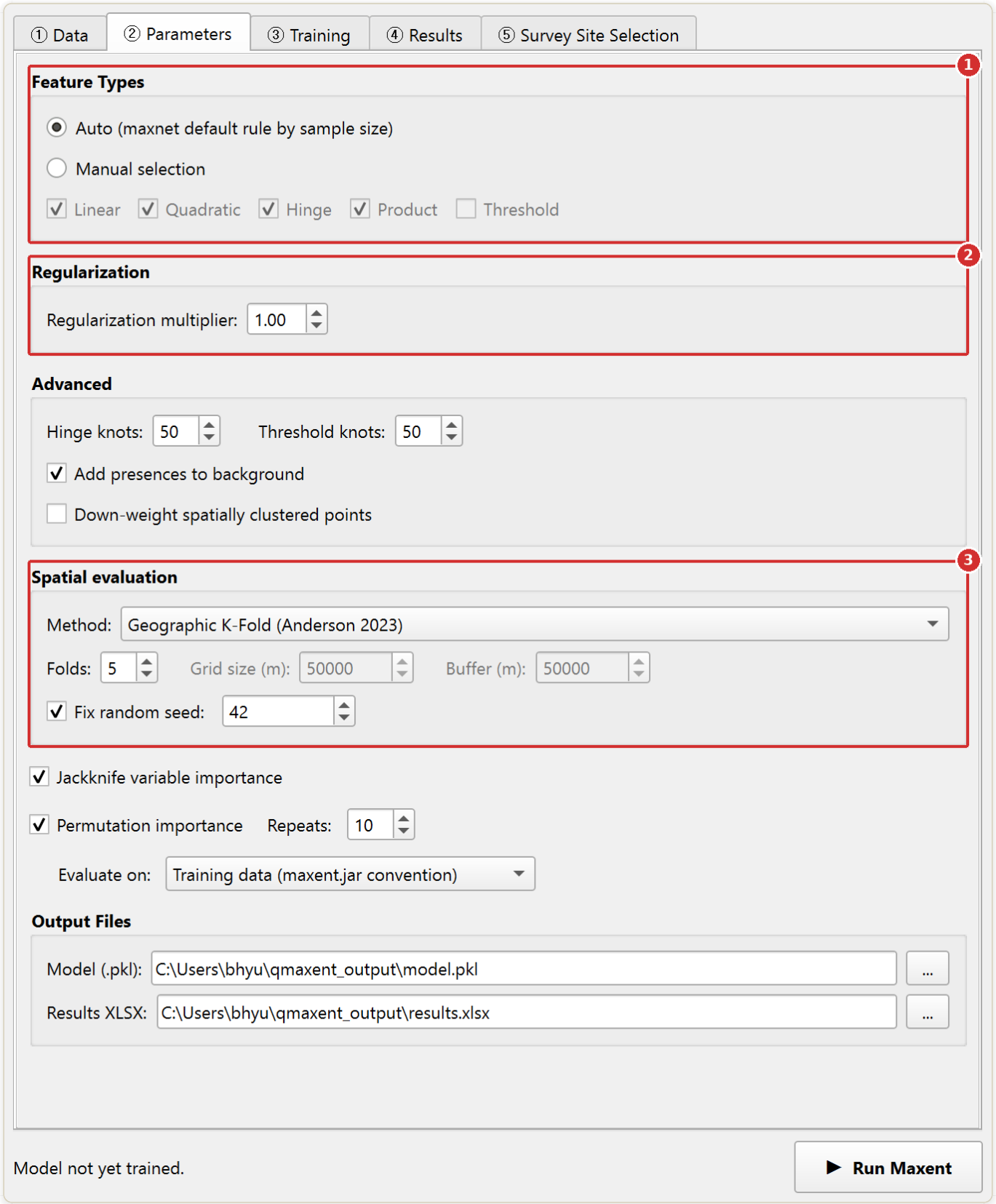
1. Click **Check Raster Consistency**.
2. The status line turns green: `✓ All 9 rasters share grid (CRS: EPSG:4326, resolution: 0.5 × 0.5)`. If it turns red instead, see Fixing rasters that do not line up.

6.3 Part 3. Fit the model (② Parameters and ③ Training)

Step 8. Review the parameters

Click the ② **Parameters** tab. For this first run, keep the defaults.

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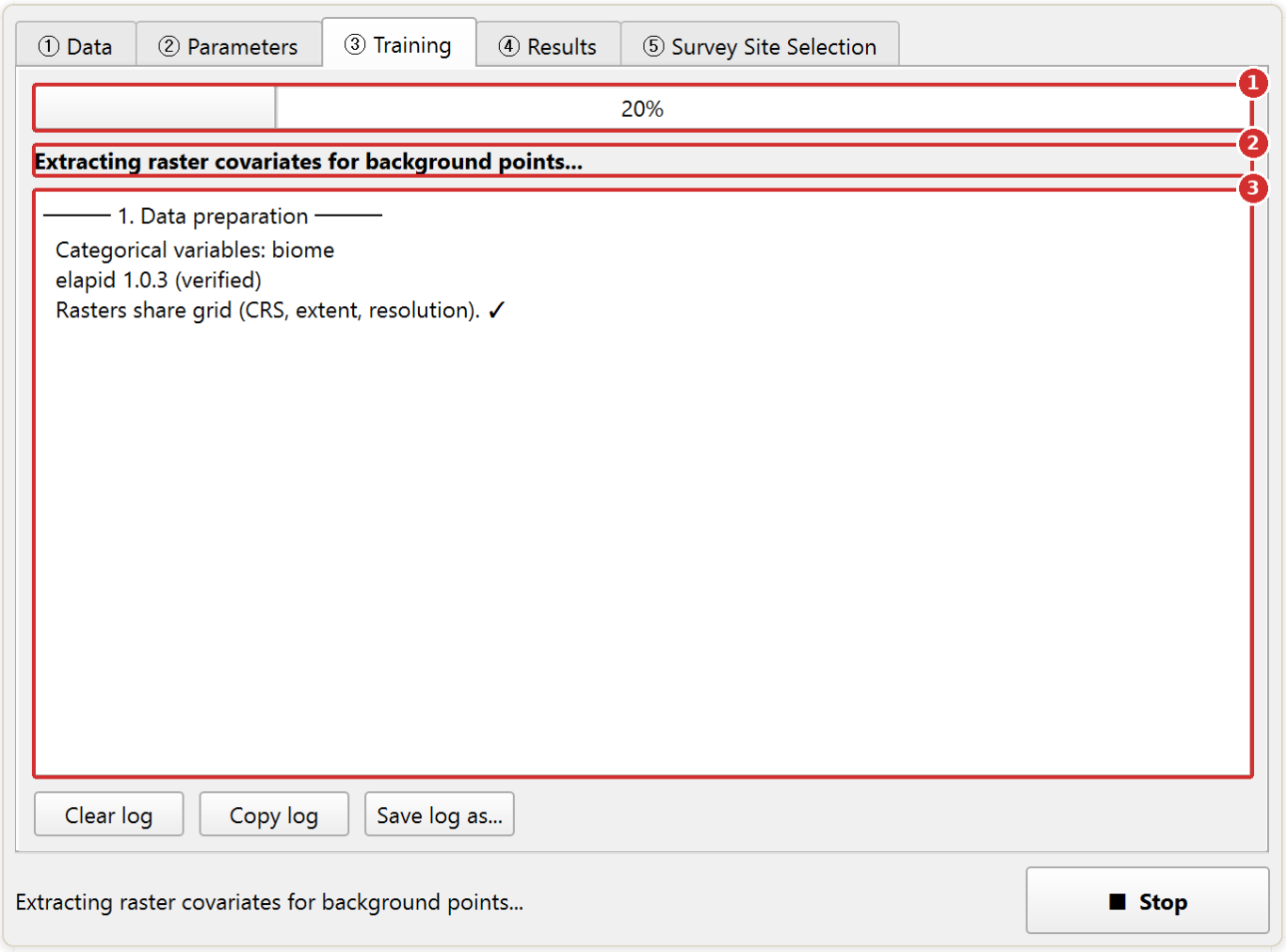
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1. **Feature Types**: **Auto (maxnet default rule by sample size)** chooses the feature classes from the number of presences. With 116 records it uses linear, quadratic, product and hinge (LQPH).
2. **Regularization**: a multiplier of 1.00 is the usual starting value.
3. **Spatial evaluation**: **Geographic K-Fold** with 5 folds checks the model on areas it did not see during fitting. **Fix random seed: 42** makes the run repeatable.

These defaults are a sensible starting point (Phillips & Dudík 2008; Phillips et al. 2017), not a guarantee of the best model for your own data.

Step 9. Run Maxent

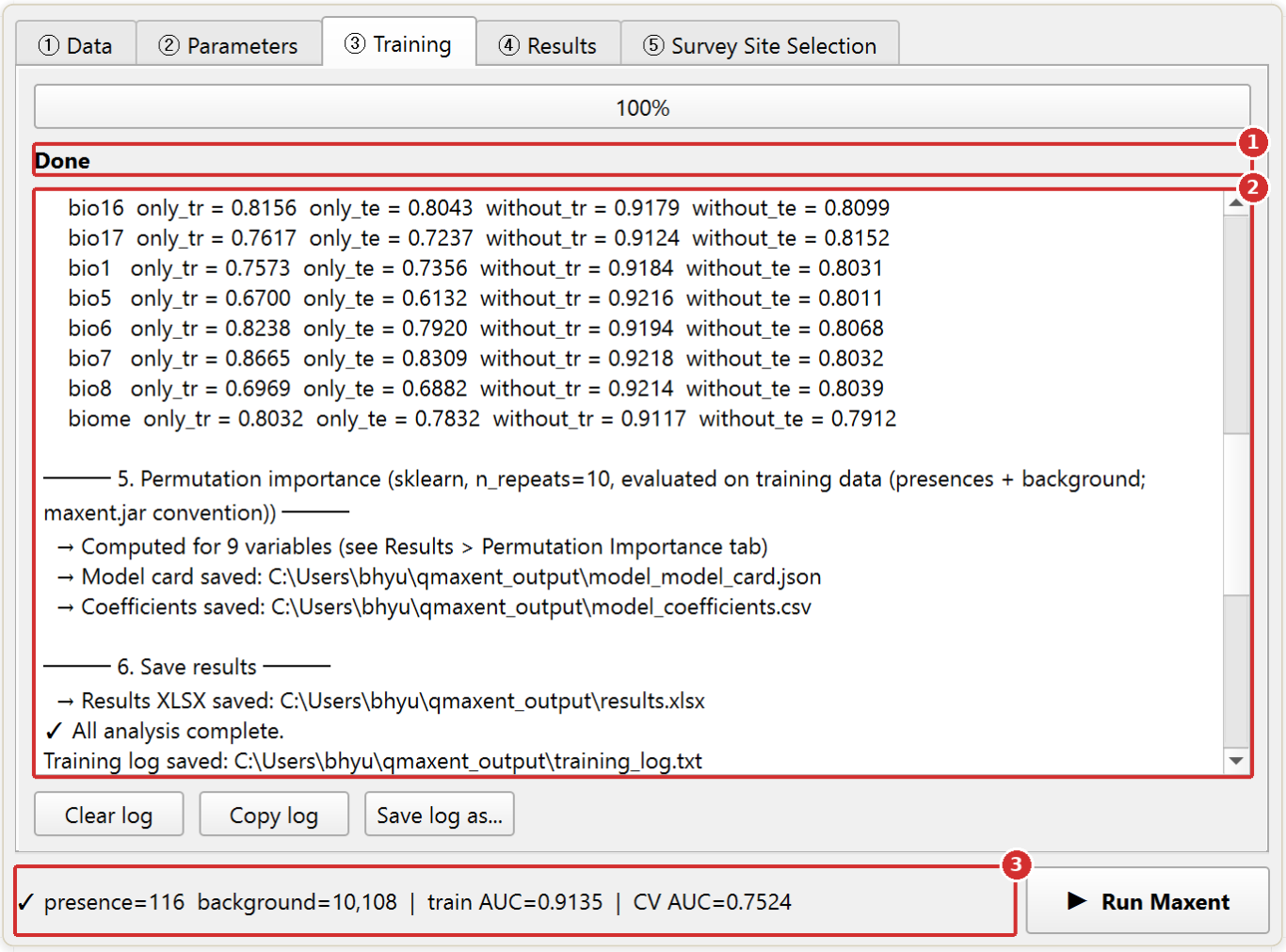
Click ▶ **Run Maxent** at the bottom of the dock. QMaxent switches to the ③ **Training** tab.



1. The progress bar shows how far the run is.
2. The line under it names the current step.
3. The log records every step as it happens.

When the run ends, the tab looks like this:

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1. The status line reports the result.
2. The log ends with a summary of the model and the evaluation.
3. The bar at the bottom of the dock keeps the key numbers in view, for example `presence=116 background=10,108 | train AUC=0.9135 | CV AUC=0.7524`.

> The test AUC from spatial cross-validation is lower than the training AUC. That gap is normal and honest: it shows how well the model does in places it was not fitted to (Roberts et al. 2017).

6.4 Part 4. Read the results and make the map (④ Results)

Step 10. Look at the response curves

Click the ④ **Results** tab.

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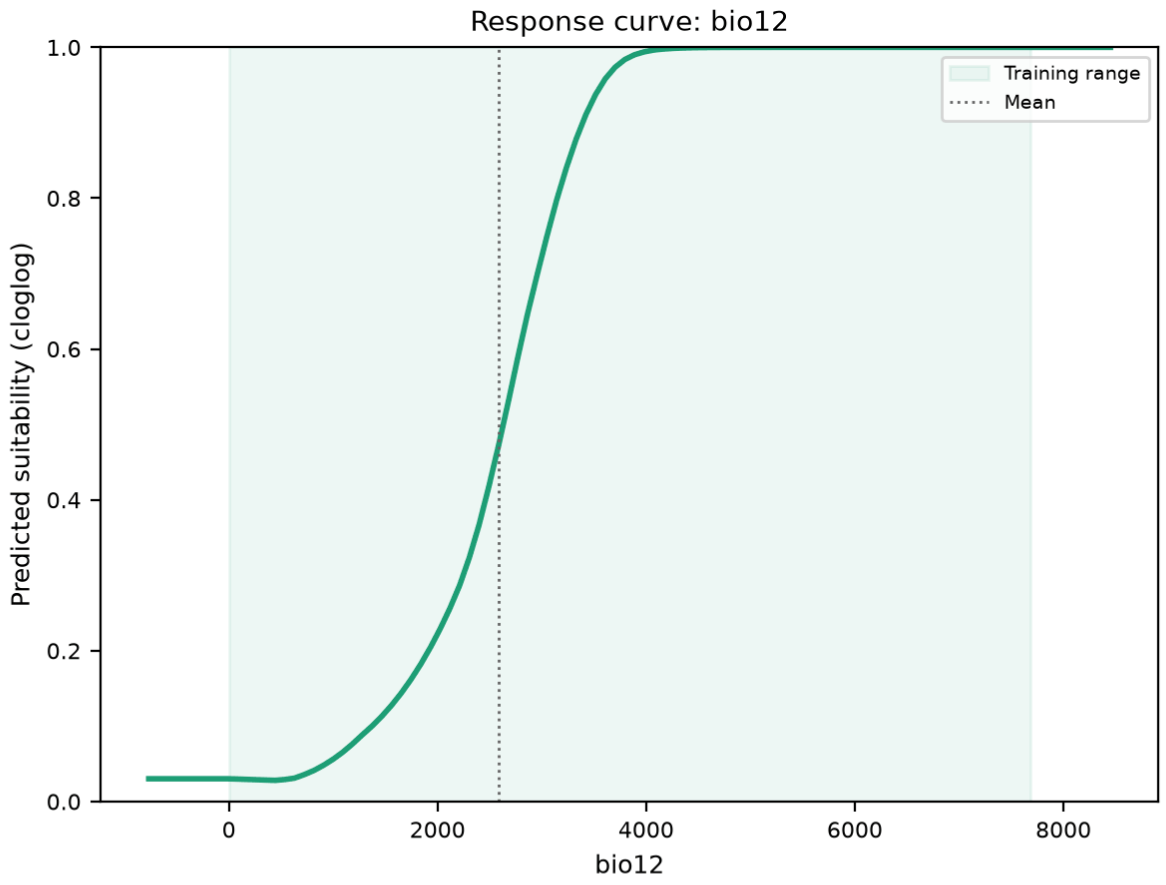
① Data | ② Parameters | ③ Training | ④ Results | ⑤ Survey Site Selection

1 Response Curves | Jackknife Importance | Permutation Importance | Spatial Projection

Marginal effect: how the predicted suitability changes across the range of one variable while the other variables are held at their training-set means. The Y-axis reads as relative habitat suitability (cloglog: 0 to 1, saturates near 1 in the most suitable region). The shaded green band marks the variable's training-data range — curves outside it are extrapolation and should be interpreted with caution; the dotted line marks the training-data mean. Useful for inspecting whether the model has learned ecologically plausible shapes (e.g. unimodal temperature optima) and for spotting variables whose curves are nearly flat — candidates for removal in a parsimonious model.

Variable: bio12 2

3



✓ presence=116 background=10,108 | train AUC=0.9135 | CV AUC=0.7524

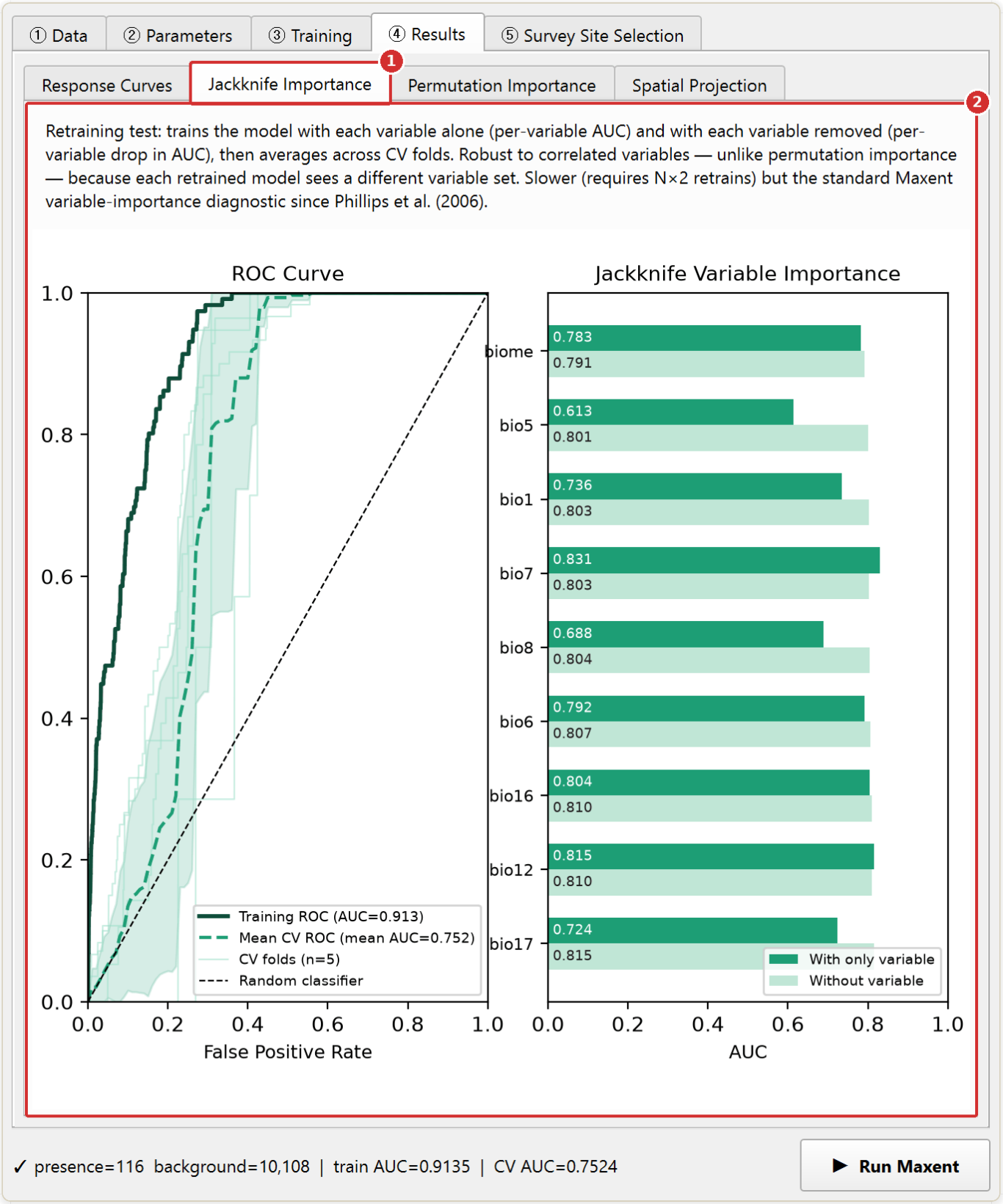
▶ Run Maxent

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1. The **Response Curves** sub-tab is open.
2. Choose a variable in **Variable**.
3. The curve shows how predicted suitability changes along that variable. The shaded band is the range of values at the training points.

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Step 11. See which variables matter



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1. Click **Jackknife Importance**.
2. The left panel shows the ROC curves. The right panel compares the model fitted with only one variable and without that variable.

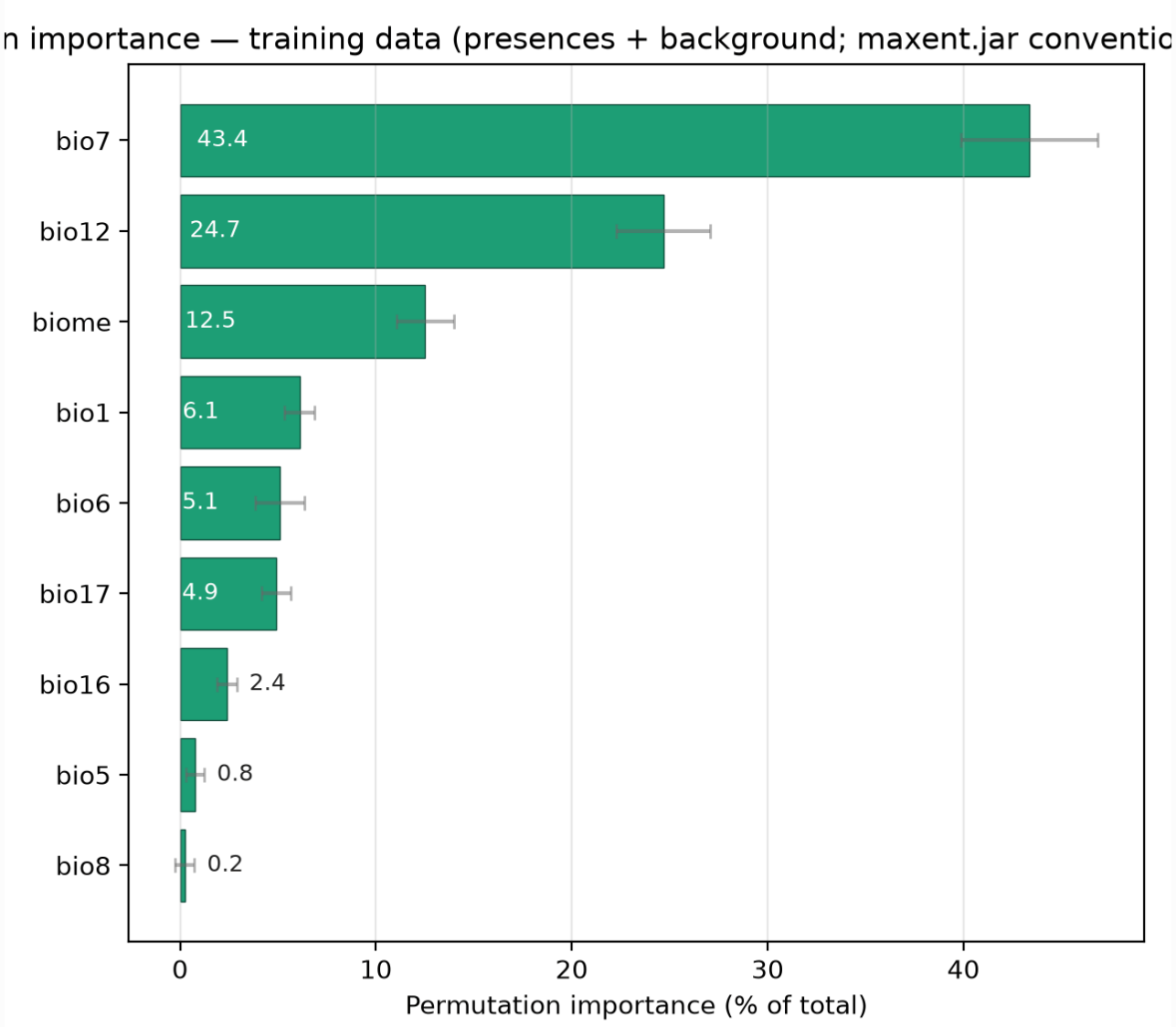
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① Data | ② Parameters | ③ Training | ④ Results | ⑤ Survey Site Selection

Response Curves | Jackknife Importance | Permutation Importance | Spatial Projection

1

Model reliance: how much the trained model's predictions depend on each variable, measured by the AUC drop after shuffling that variable's values. Same metric reported by maxent.jar as "permutation importance". Fast to compute but can underestimate the importance of variables correlated with other variables in the model (Strobl et al. 2007); read alongside the Jackknife tab rather than alone.



2 Save PNG... 3 Save CSV...

✓ presence=116 background=10,108 | train AUC=0.9135 | CV AUC=0.7524

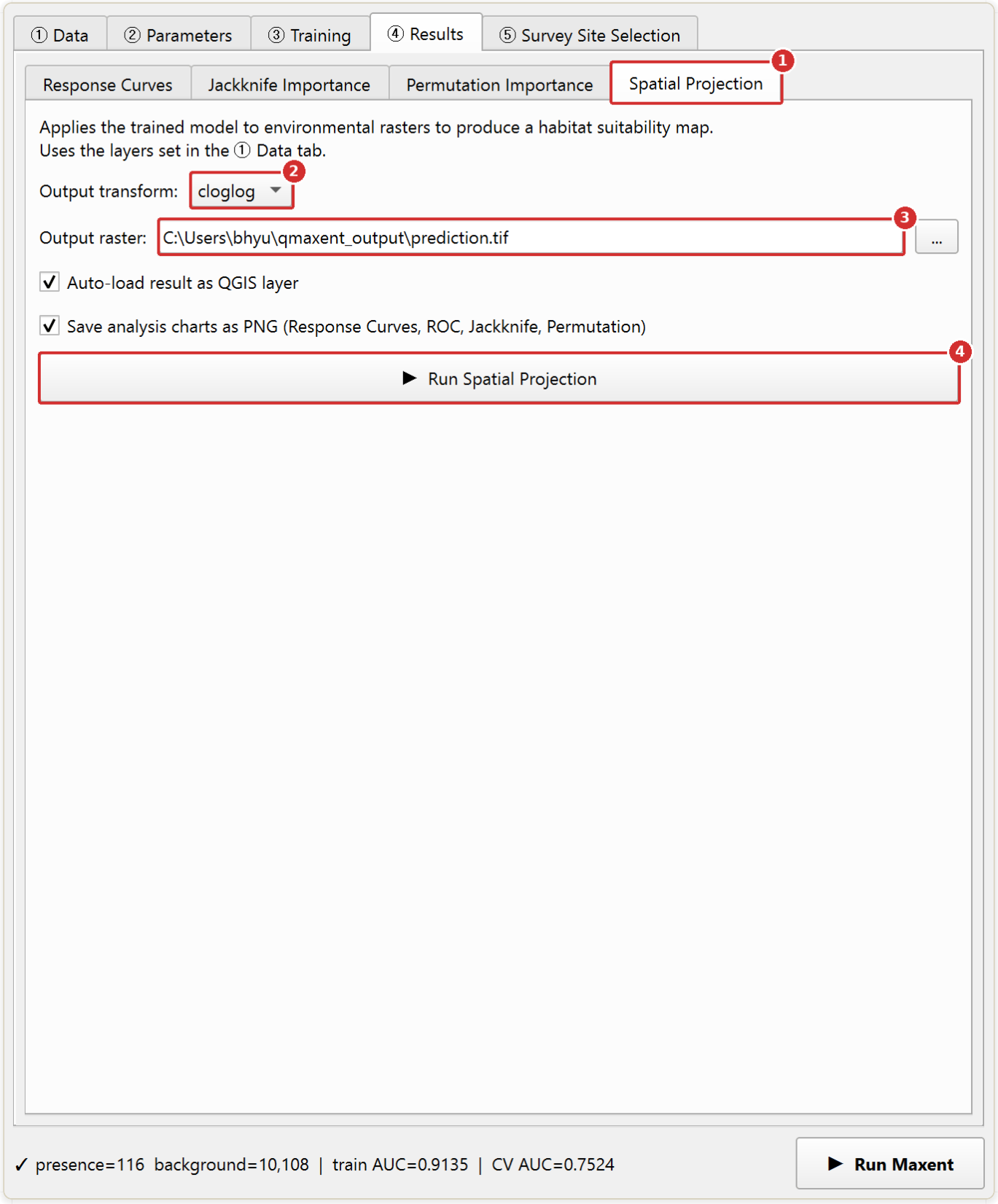
▶ Run Maxent

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1. **Permutation Importance** ranks the variables by how much the model gets worse when each one is shuffled.
2. **Save PNG…** saves the chart.
3. **Save CSV…** saves the numbers.

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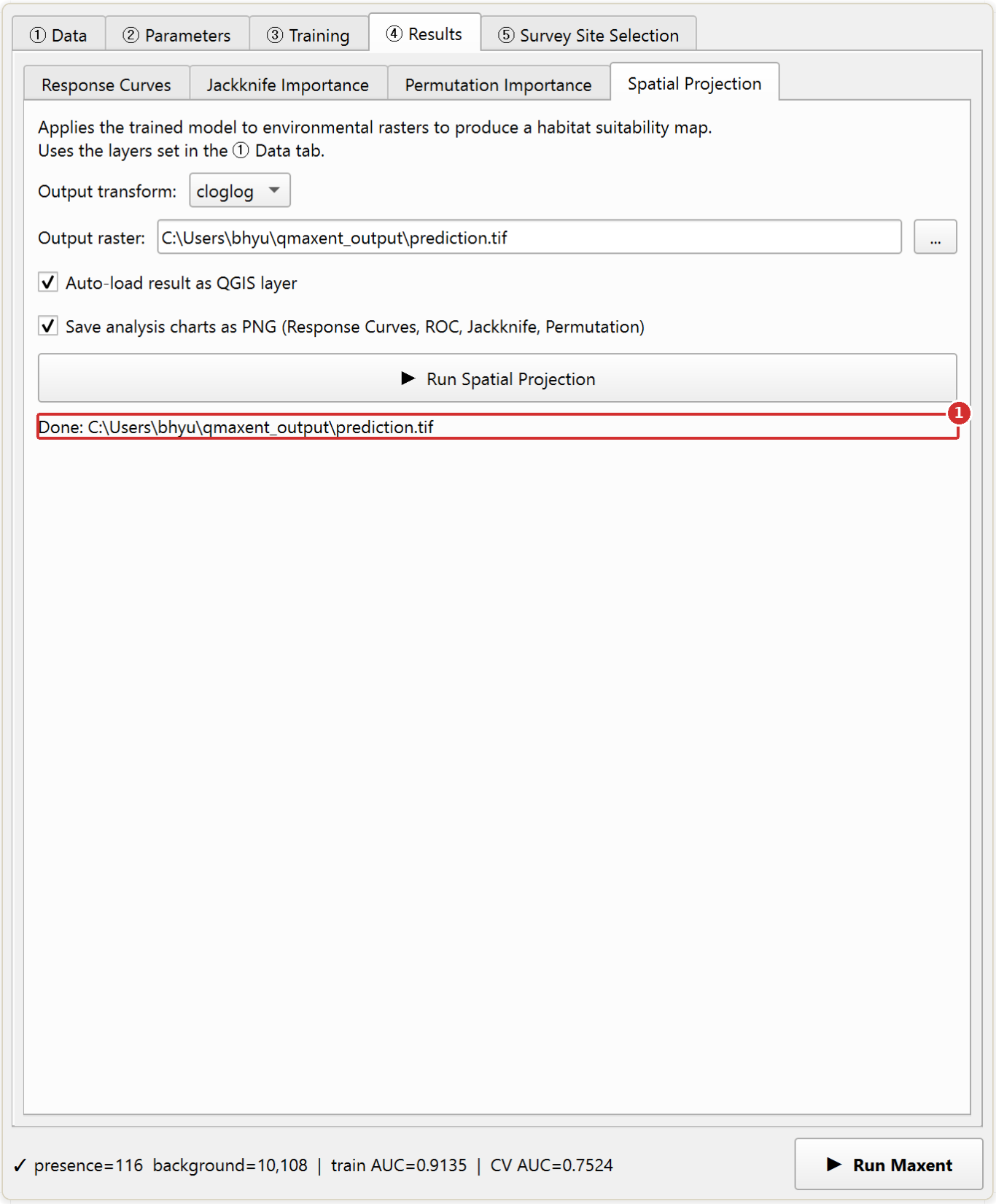
Step 12. Make the suitability map



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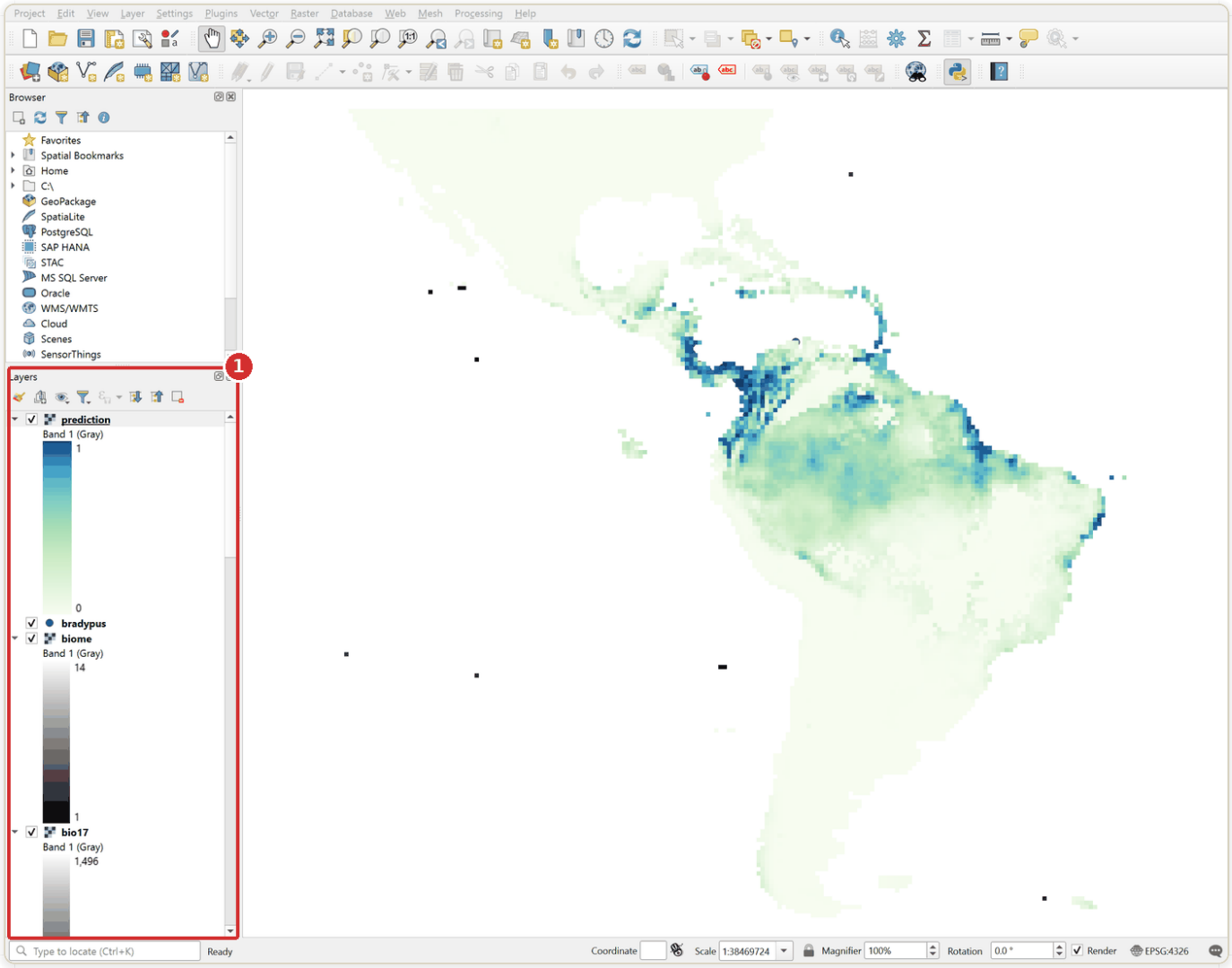
1. Click **Spatial Projection**.
2. Keep **cloglog** in **Output transform**. It gives values from 0 to 1 that read as relative suitability (Phillips et al. 2017).
3. **Output raster** is where the map is saved. Click **…** to change it.
4. Click ▶ **Run Spatial Projection**.

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1. When the status line reads `Done:` followed by the file path, the map is saved and added to the project.

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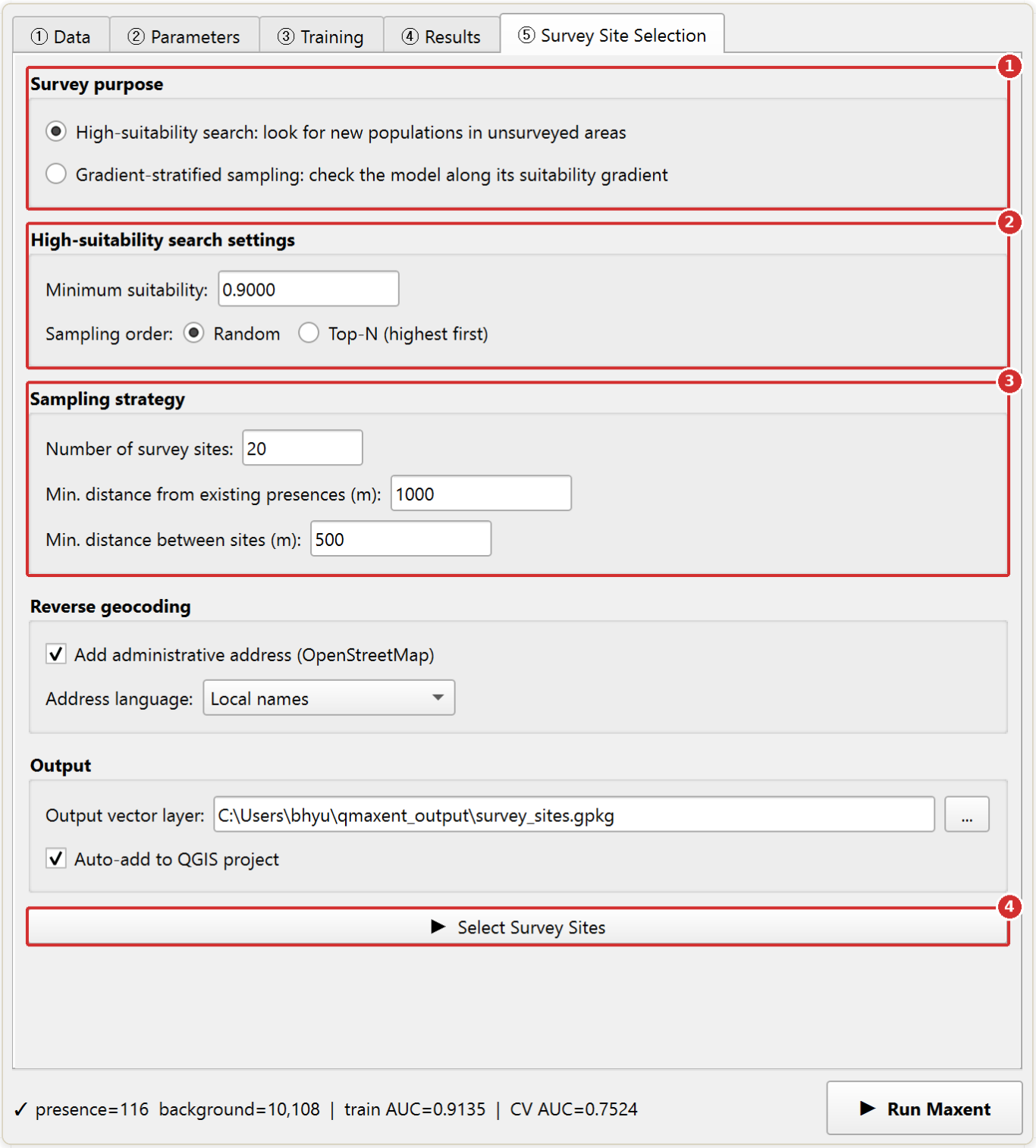
1. The `prediction` layer appears in the **Layers** panel. Darker colours mean higher predicted suitability.

6.5 Part 5. Pick survey sites (⑤ Survey Site Selection)

Step 13. Set up the survey

Click the ⑤ **Survey Site Selection** tab.

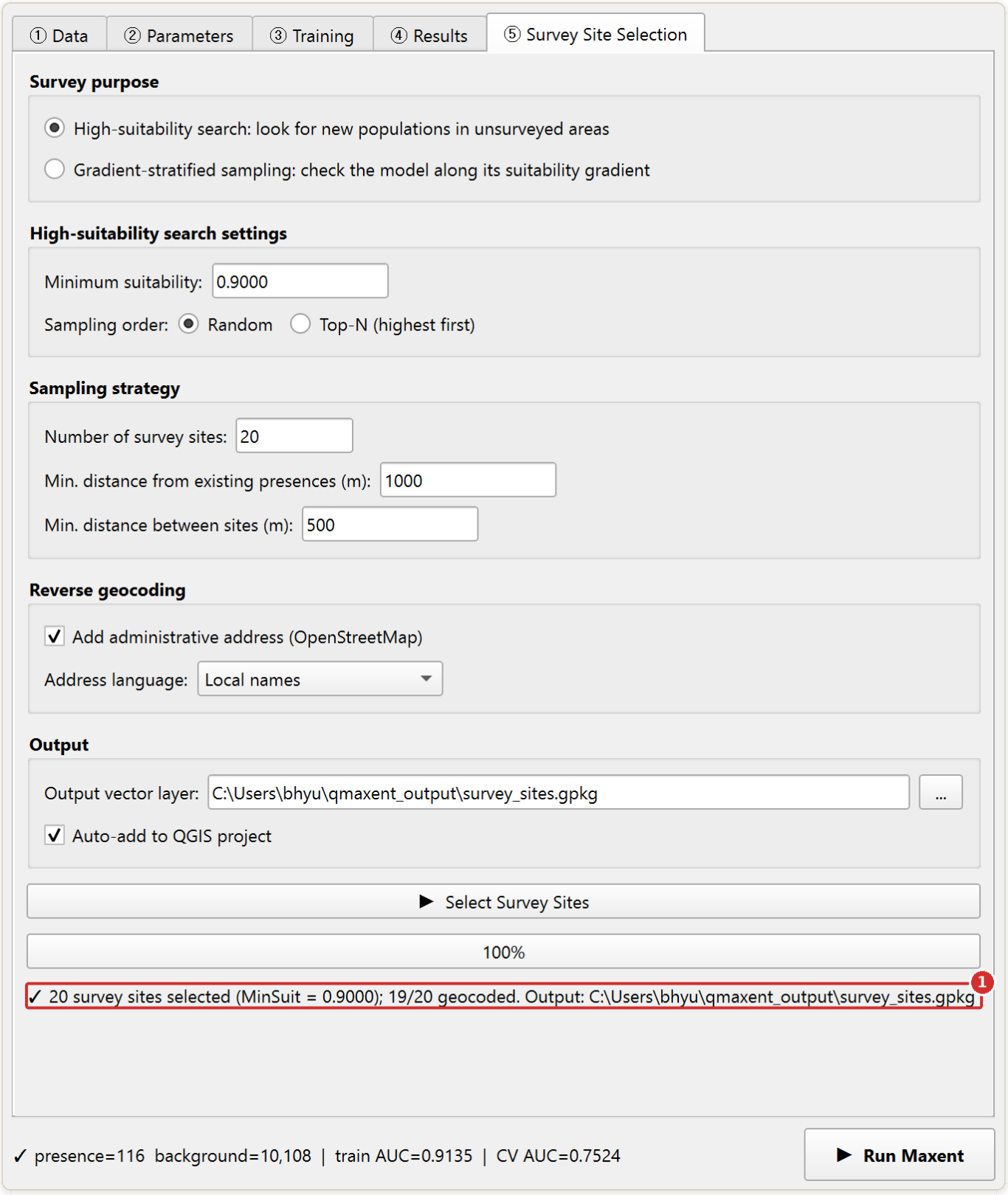
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1. **Survey purpose**: choose **High-suitability search** to look for new populations in places that are suitable but not yet surveyed.
2. **High-suitability search settings**: **Minimum suitability** is the lowest map value a site may have.
3. **Sampling strategy**: set how many sites you want and how far apart they must be from known records and from each other (in metres).
4. Click ▶ **Select Survey Sites**.

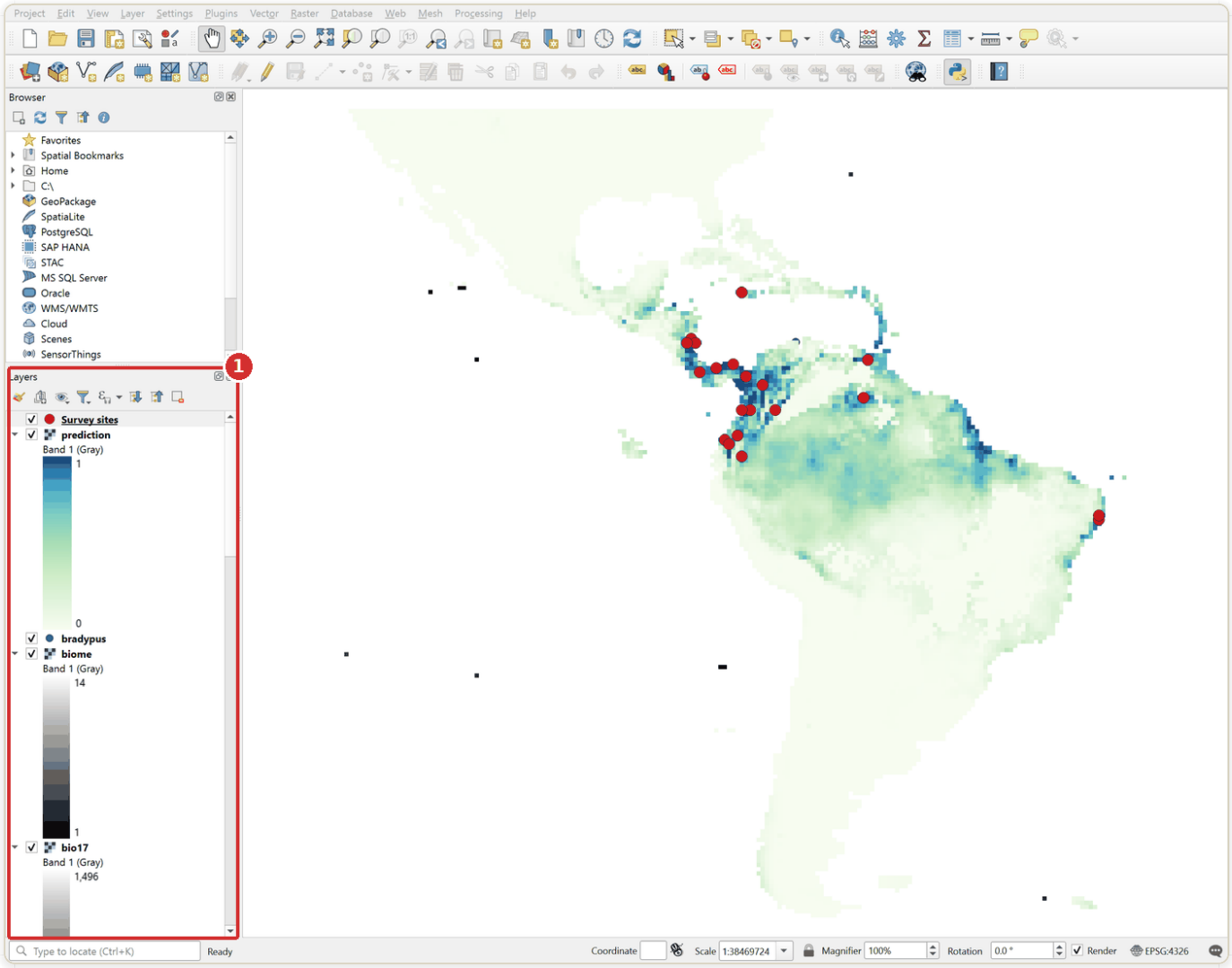
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Step 14. Check the survey sites



1. The status line reports the result, for example `✓ 20 survey sites selected (MinSuit = 0.9000)`, and where the GeoPackage was saved.

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1. The `Survey sites` layer is added on top of the map. Open its attribute table to see the coordinates, the suitability value and, if reverse geocoding was on, the address of each site.

6.6 What you have done

In these steps you ran a complete species distribution modelling workflow inside QGIS. You loaded presence points and environmental rasters, checked that the rasters line up, fitted a regularized Maxent model, evaluated it with spatial cross-validation, read the response curves and variable importance, mapped habitat suitability and turned the map into a list of field survey sites.

6.7 Next steps

- Read the **User guide** chapters to learn what each control does and when to change its default.
- Work through the Bradypus worked example for a closer reading of every plot.
- Practise fixing rasters that do not line up with the Ariolimax example.

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- Reproduce a published study with the Pitta nympha example.

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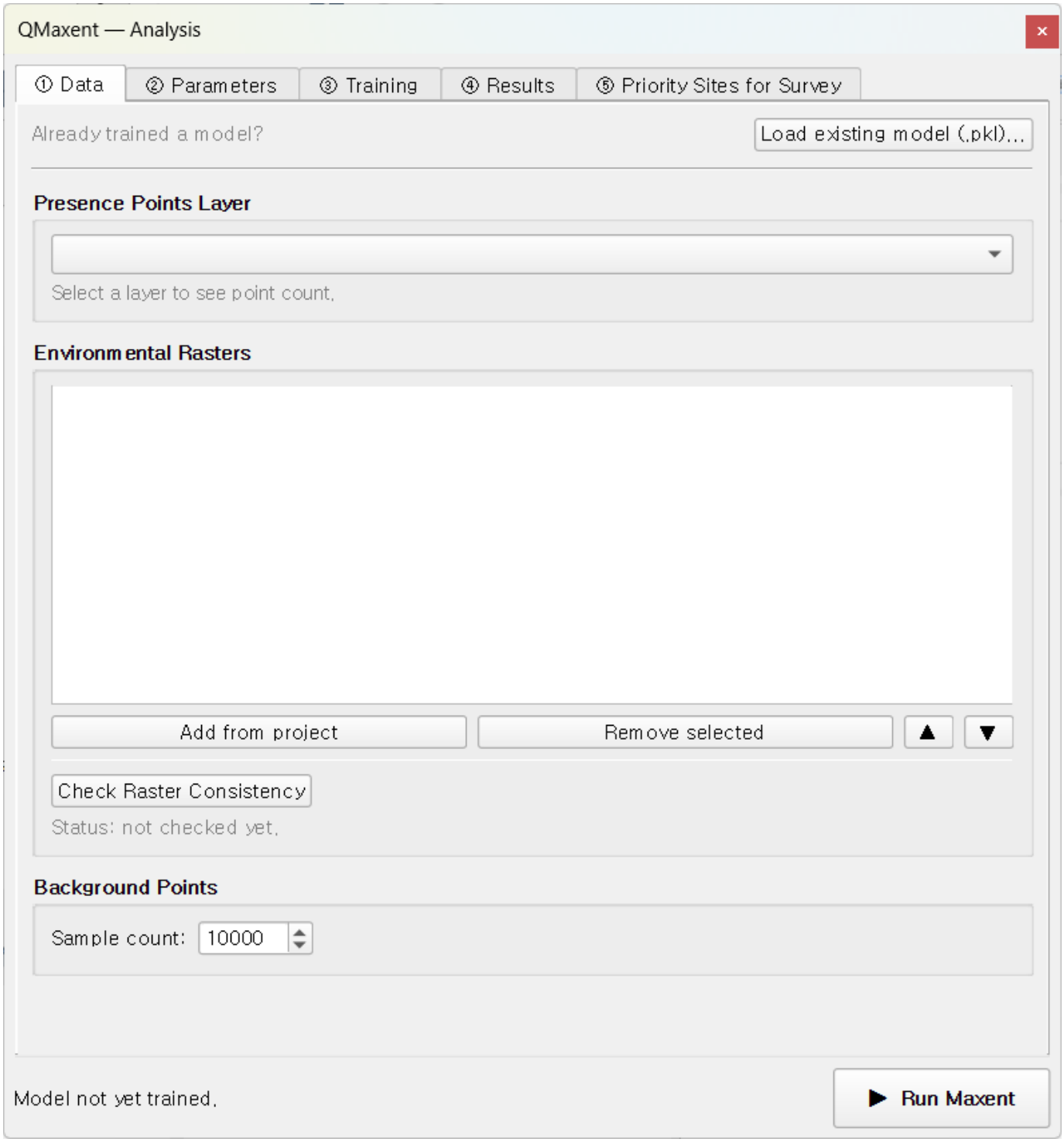
7. The Analysis dock

The QMaxent Analysis dock is the heart of the plugin. It organises the entire SDM workflow into **five numbered tabs** that you progress through in order, left to right. This chapter is a quick tour of the layout; each tab then has its own detailed chapter.

7.1 Opening the dock

Choose **Plugins → QMaxent → QMaxent Analysis**. The dock opens on the right side of the QGIS main window by default and can be detached, floated, or re-docked using the standard QGIS panel handles.

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7.2 The five tabs at a glance

#	Tab	Purpose	Detailed chapter
①	**Data**	Pick or load presence points (point file or CSV); register environmental rasters; pre-flight raster grid and CRS check	① Data tab
②	**Parameters**	Choose feature classes, regularization, spatial CV scheme, output paths	② Parameters tab
③	**Training**	Live progress bar and log while the model fits	③ Training tab
④	**Results**	Response curves, Jackknife importance, spatial projection	④ Results tab
⑤	**Survey Site Selection**	Turn the prediction into field-ready candidate survey sites	⑤ Survey Site Selection

The numbered glyphs are deliberate: even when QGIS is configured in a language other than English, the order is unambiguous.

7.3 Status bar

A persistent **status bar at the bottom of the dock** shows the most recent run summary — presence count, background count, training AUC, and CV AUC. It is updated continuously as you click around, so you can switch projects or reopen QGIS and immediately see whether the dock is showing a fresh state or a previous run's results.

7.4 Persistence across QGIS sessions

QMaxent remembers your last-used presence layer, raster set, parameter values, and output paths inside the QGIS project file (`.qgz`). When you reopen a project, the dock reflects the previous state — no need to re-pick layers each time.

7.5 Buttons across the bottom

The footer of the dock has three persistent buttons:

- ▶ **Run Maxent** — start a training run
- ▶ **Run Spatial Projection** — apply the trained model across all rasters
- ▶ **Select Survey Sites**: sample candidate survey sites from the prediction

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These buttons are **enabled only when their prerequisites are met**: e.g. projection becomes available only after a model has been trained, and survey site selection requires a projection raster on disk.

7.6 What this means for your workflow

The dock's design encodes the canonical Maxent workflow described in Elith et al. 2011: assemble inputs → choose hyperparameters → fit → evaluate → project → act. By making each stage a separate tab with its own status, QMaxent forces a discipline that helps catch the silent failures Roberts et al. 2017 and Araújo et al. 2019 warn about — particularly raster mismatch and over-optimistic AUCs from non-spatial CV.

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8. ① Data tab

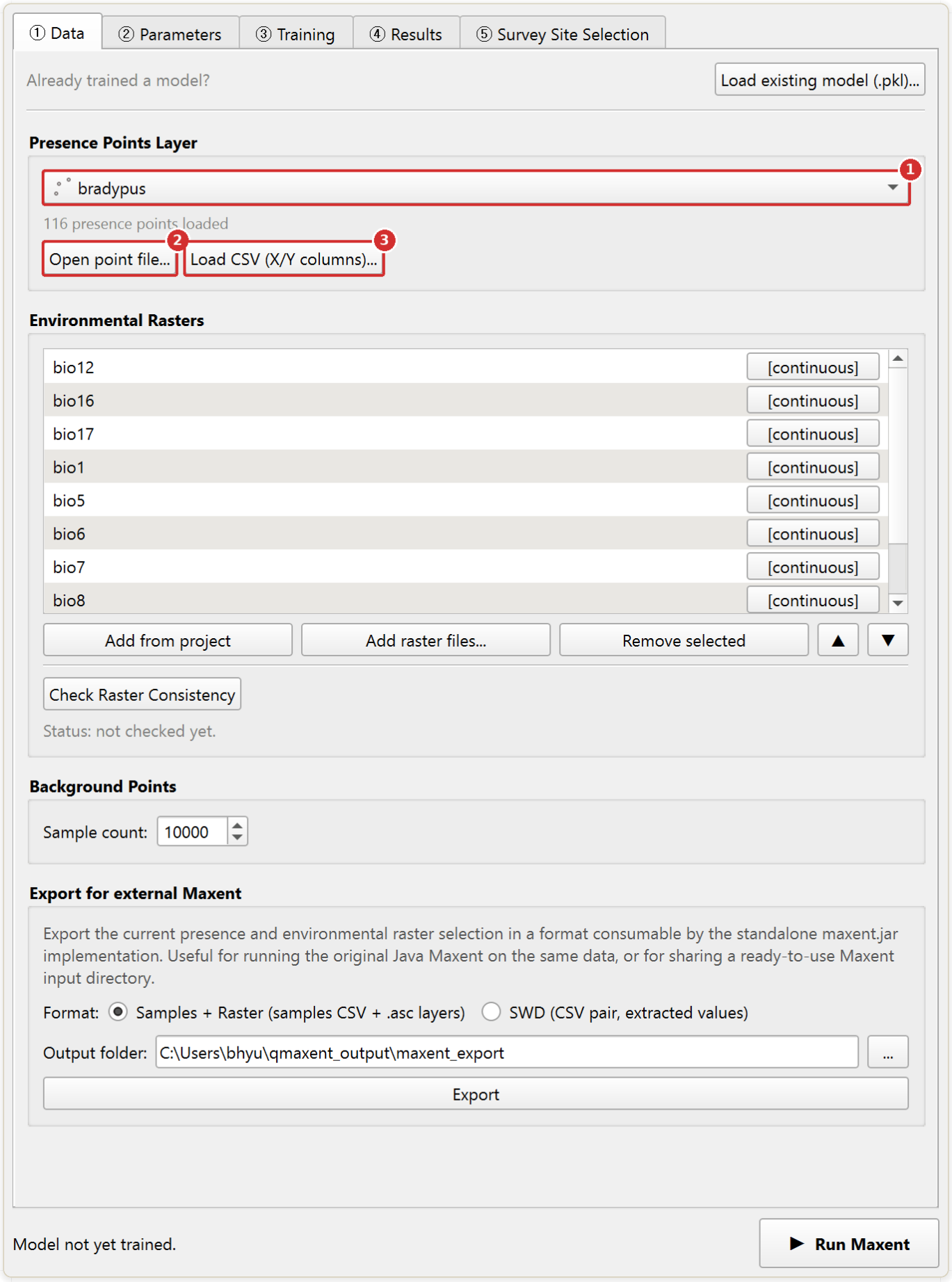
The Data tab is where you tell QMaxent **which species** you are modeling and **which environmental variables** to use. Three controls deserve attention: the presence-points selector, the environmental-raster list with its **Check Raster Consistency** action, and the background-points sample size.

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8.1 Step by step

Step 1. Choose the presence points

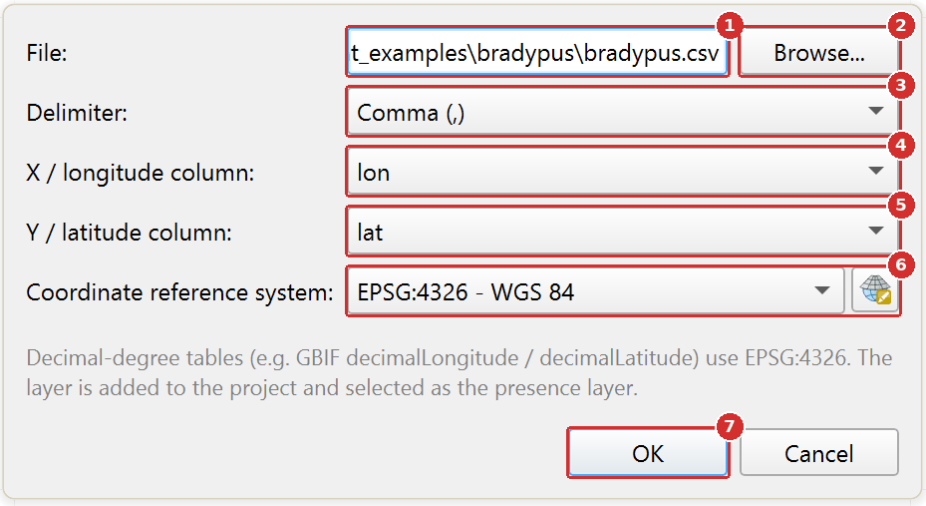
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1. **Presence Points Layer** lists every point layer in the project. Choose your species layer. The grey line below it tells you how many points were loaded.
2. **Open point file…** adds a point file (GeoPackage, shapefile, GeoJSON and other formats QGIS reads) to the project and selects it.
3. **Load CSV (X/Y columns)…** reads a table with coordinate columns (see Step 2).

Step 2. (Optional) Load presence points from a CSV table

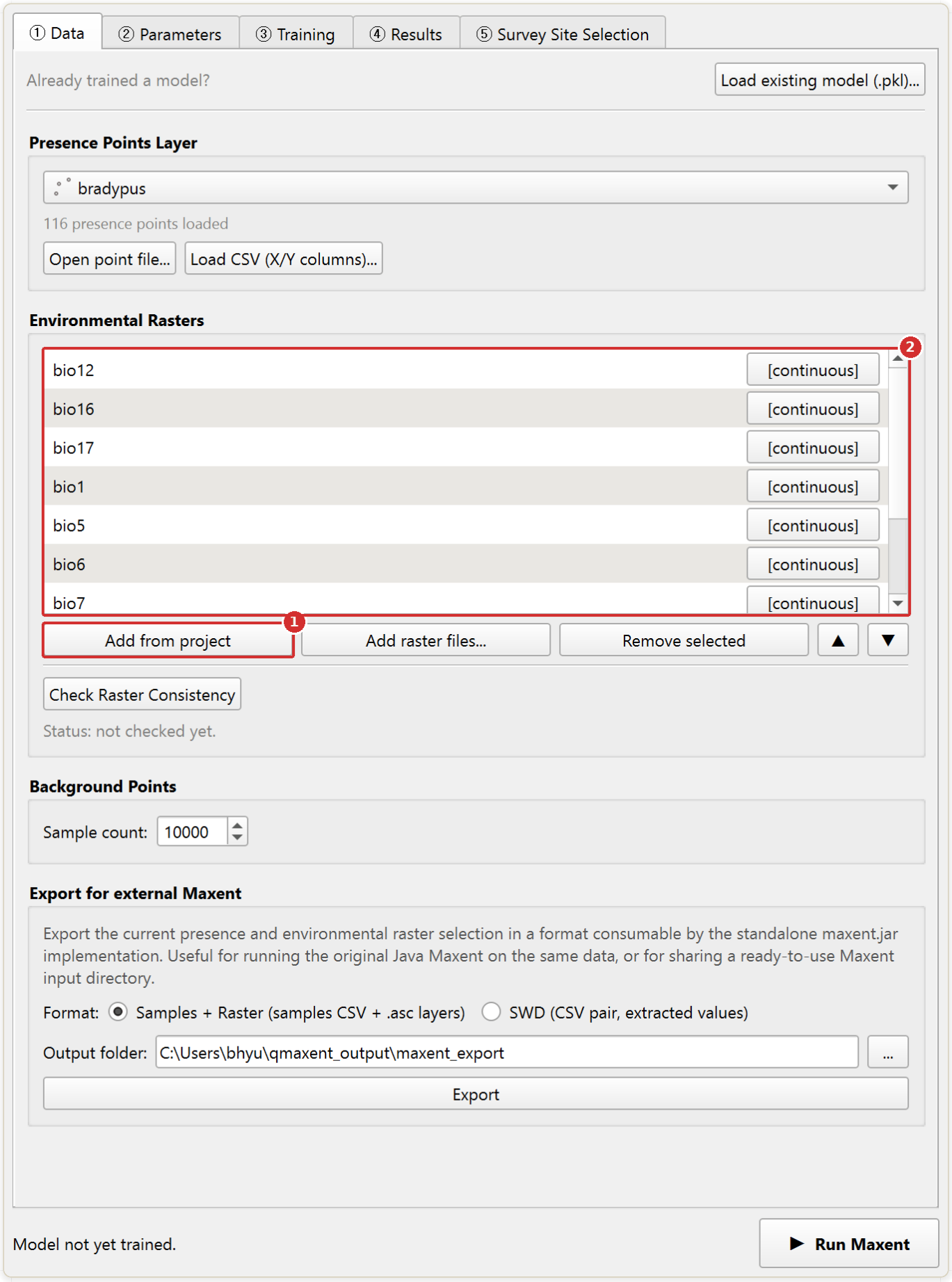


1. **File**: the path of the table.
2. **Browse…** chooses the file.
3. **Delimiter** is detected for you. Change it only if the columns look wrong.
4. **X / longitude column**: the longitude or easting column.
5. **Y / latitude column**: the latitude or northing column. QMaxent guesses both from the column names (for example `lon` / `lat`, `x` / `y`).
6. **Coordinate reference system**: the CRS of the numbers in the table. Use EPSG:4326 for decimal degrees.
7. Click **OK**. The table is added to the project as a point layer.

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Step 3. Add the environmental rasters

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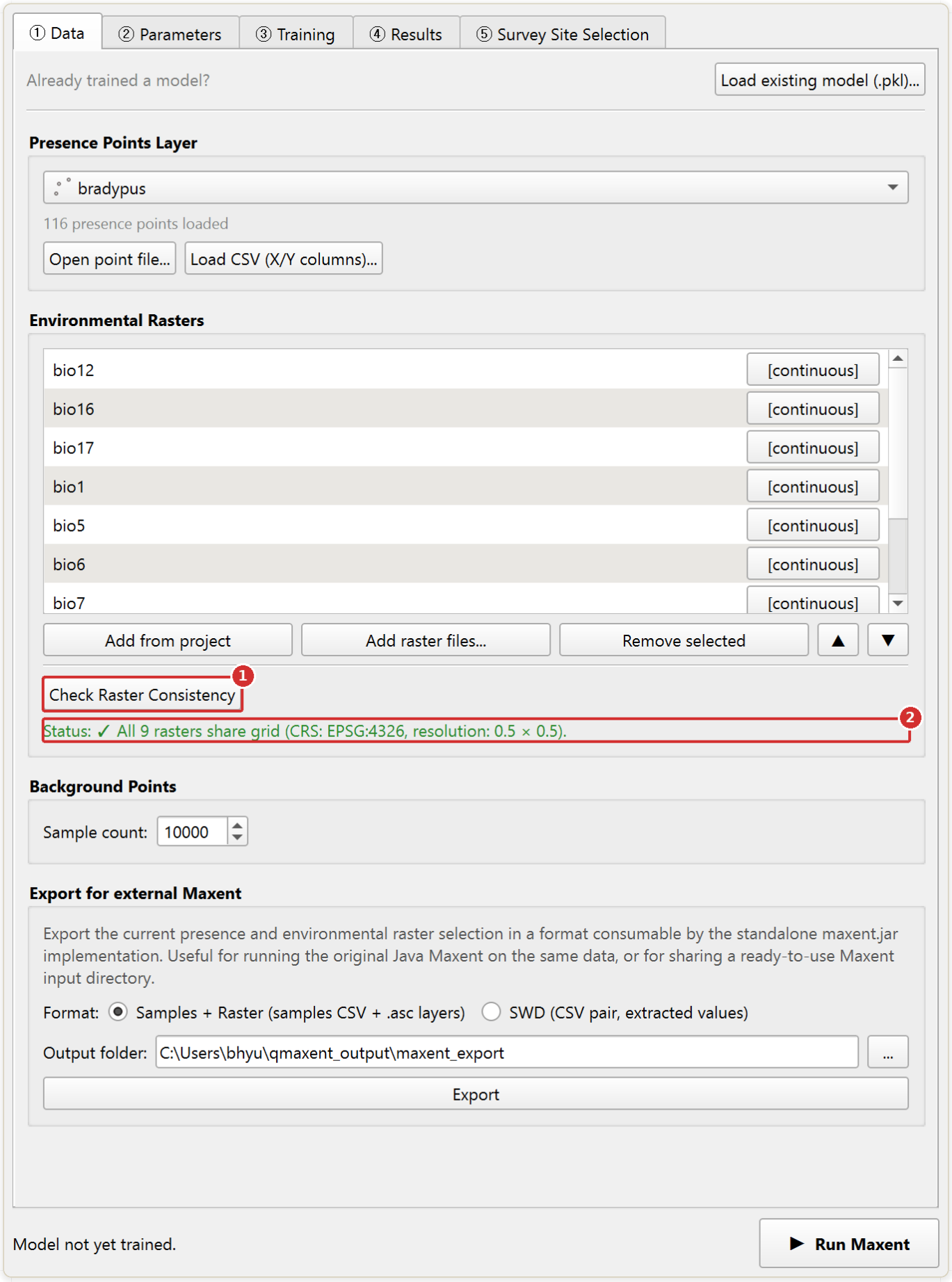
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1. **Add from project** puts every raster layer of the project in the list. **Add raster files…** adds rasters straight from disk.
2. Each row has a type button. `[continuous]` is for measured amounts such as temperature; `[categorical]` is for class codes such as land cover or biome. Click the button to switch. Use **Remove selected** and the ▲ ▼ buttons to remove rows or change their order.

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Step 4. Check that the rasters line up

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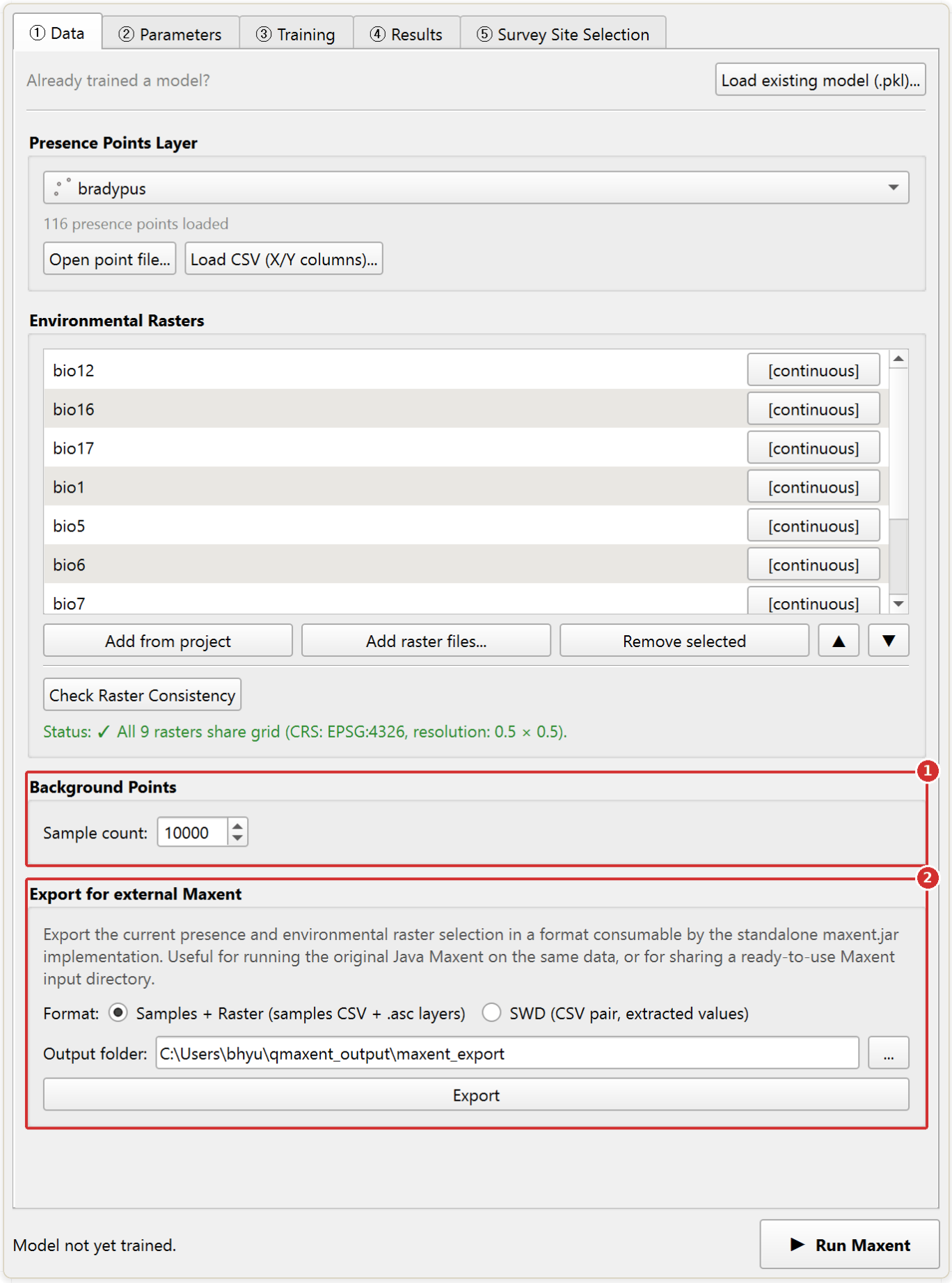
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1. Click **Check Raster Consistency**.
2. A green status line means every raster has the same CRS, extent and resolution. A red line means they differ; see Fixing rasters that do not line up.

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Step 5. Check the background points

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1. **Background Points**: **Sample count** is the number of random background points (10,000 by default). See Background points below.
2. **Export for external Maxent** is optional. It writes the same data for the Java Maxent program (see Export for external Maxent).

When these steps are done, go to the ② Parameters tab.

8.2 Presence Points Layer

The first drop-down lists every loaded vector point layer in the current QGIS project. Pick the layer that contains your species occurrence records. QMaxent reads the geometry and reports the count beside the selector (e.g. `116 presence points loaded`).

If the records are not in the project yet, two buttons below the drop-down load them without a detour through the QGIS menus. The new layer is added to the project and selected as the presence layer.

- **Open point file…**: any point format QGIS can read (GeoPackage, Shapefile, GeoJSON, KML, GML, FlatGeobuf and others). A file that does not contain points is refused with a message.
- **Load CSV (X/Y columns)…**: a delimited table with one row per record, for example a GBIF download. QMaxent detects the delimiter (comma, semicolon, tab or pipe) and suggests the longitude/latitude or X/Y columns from their names (for example `decimalLongitude` and `decimalLatitude` in GBIF files, or `lon`/`lat`, `x`/`y`). Check the suggested columns and choose the coordinate reference system of the coordinates. The default is EPSG:4326, which is correct for decimal-degree tables.

Tips for preparing this layer:

- **Geometry** must be `Point` or `MultiPoint`. Lines and polygons must be converted to centroids first via **Vector → Geometry Tools → Centroids…**.
- **CRS** can be anything — QMaxent reprojects on the fly to match the raster grid. That said, a projected CRS that matches the rasters avoids a mid-run reprojection step.
- **Duplicates and clustering** are tolerated but distort the model. maxent.jar keeps one presence per raster cell by default; QMaxent keeps all records, and the ③ Training tab log reports how many share a cell. Reduce the records to one per cell or apply **spatial thinning** before modeling (Boria et al. 2014), for example with the standalone tool *spThin*. Records with identical coordinates can also be removed with `Vector → Geoprocessing Tools → Delete Duplicate Geometries`.
- **Very large presence sets** (hundreds of thousands of points) are usually dominated by such duplicates and need several GB of memory per model fit (about 4 GB for 100,000 points in our tests). Thin them first; the model rarely changes once each occupied cell is represented once.

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8.3 Environmental rasters

Two buttons populate the raster list:

- **Add from project**: adds every raster layer currently loaded in the QGIS project. The fastest way to start.
- **Add raster files…**: opens a multi-select file dialog for any single-band raster GDAL can read (GeoTIFF, ESRI ASCII `.asc`, `.grd`, ERDAS `.img`, VRT, NetCDF and others). The files are added to the QGIS project and to the list in one step.

The list follows the QGIS project: when a raster layer is removed from the project (or you open another project), its row disappears from the list and the status line asks you to run **Check Raster Consistency** again. Use the ▲ and ▼ buttons to change the order of the rows. The order matters for reproducibility, because background points are drawn from the first raster in the list (see Background points).

QMaxent uses **one band per environmental variable**. A multi-band raster (for example a stack of all bioclimatic variables in one file) is refused when you click ▶ **Run Maxent**, with the names of the affected files. Split it into single-band files first, for example with **Processing > GDAL > Raster conversion > Rearrange bands** (one band per output), then add the single-band files.

Each row in the raster list shows the file name and a `[continuous]` / `[categorical]` tag. **Categorical detection is automatic** when the file carries a PAM (`.aux.xml`) sidecar with the metadata key `LAYER_TYPE=thematic`. Without that tag, click the tag button of a row to switch it between `[continuous]` and `[categorical]`.

> **Why categorical handling matters**
>
> Treating a categorical raster (e.g. biome, land-cover class) as continuous is one of the most common silent failures in operational SDM. Maxent will average across class codes as if they were ordinal, producing biologically meaningless responses. Always verify the `[categorical]` tag on integer-coded landscape rasters before training.

8.4 Check Raster Consistency

This button runs the most important pre-flight check in the dock: do all rasters share the **same CRS, extent, and resolution**?

- ✓ **Pass** (green): the stack is grid-aligned and ready to model.
- ⚠ **Mismatch** (amber): one or more rasters disagree.
- **Unusable CRS**: one or more rasters have a coordinate reference system that cannot be reprojected (see below).

When it fails, a second button — **Harmonize to Folder…** — appears. Clicking it lets you nominate an output directory; QMaxent reprojects every raster to the highest-resolution one in the stack (nearest-

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neighbour for categoricals, bilinear for continuous variables) and re-loads the harmonised set. The Ariolimax worked example walks through this in detail.

Rasters with an unusable CRS

Some rasters cannot be aligned at all because of their CRS. The check reports two cases and names each affected file:

- the raster has **no CRS**, or
- the CRS is a **local (engineering) system without a datum**. This happens when a `.prj` file or GeoTIFF only carries the name of a projection, for example `NAD83(NSRS2007) / Michigan South`, and no datum. PROJ cannot transform such a CRS to any other CRS.

When the name matches an EPSG definition, the message suggests the matching EPSG code. To fix it, in the QGIS **Layers** panel right-click the raster, choose **Layer CRS > Set Layer CRS...**, select the correct CRS (for example the suggested EPSG code), and click **Check Raster Consistency** again. From then on, Check and **Harmonize to Folder...** use the CRS you set in QGIS for that raster. Alternatively, write a complete CRS into the file, for example with **Raster > Projections > Assign Projection**. Harmonize no longer stops with a raw PROJ error in this situation.

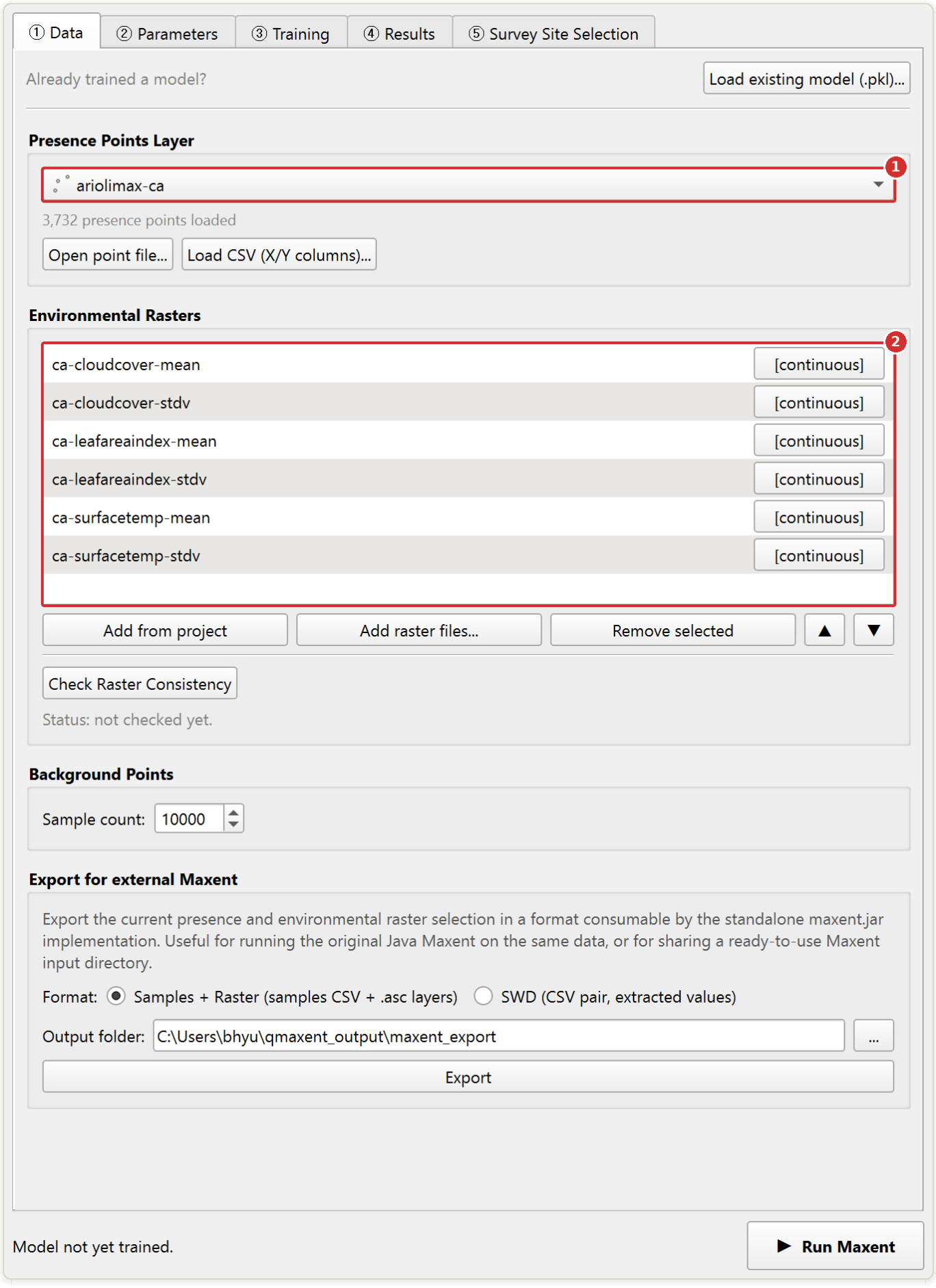
8.5 Fixing rasters that do not line up

The steps below use the Ariolimax **Mismatch demo** from Example datasets, whose rasters were changed on purpose.

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Step 1. Add the rasters and run the check

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